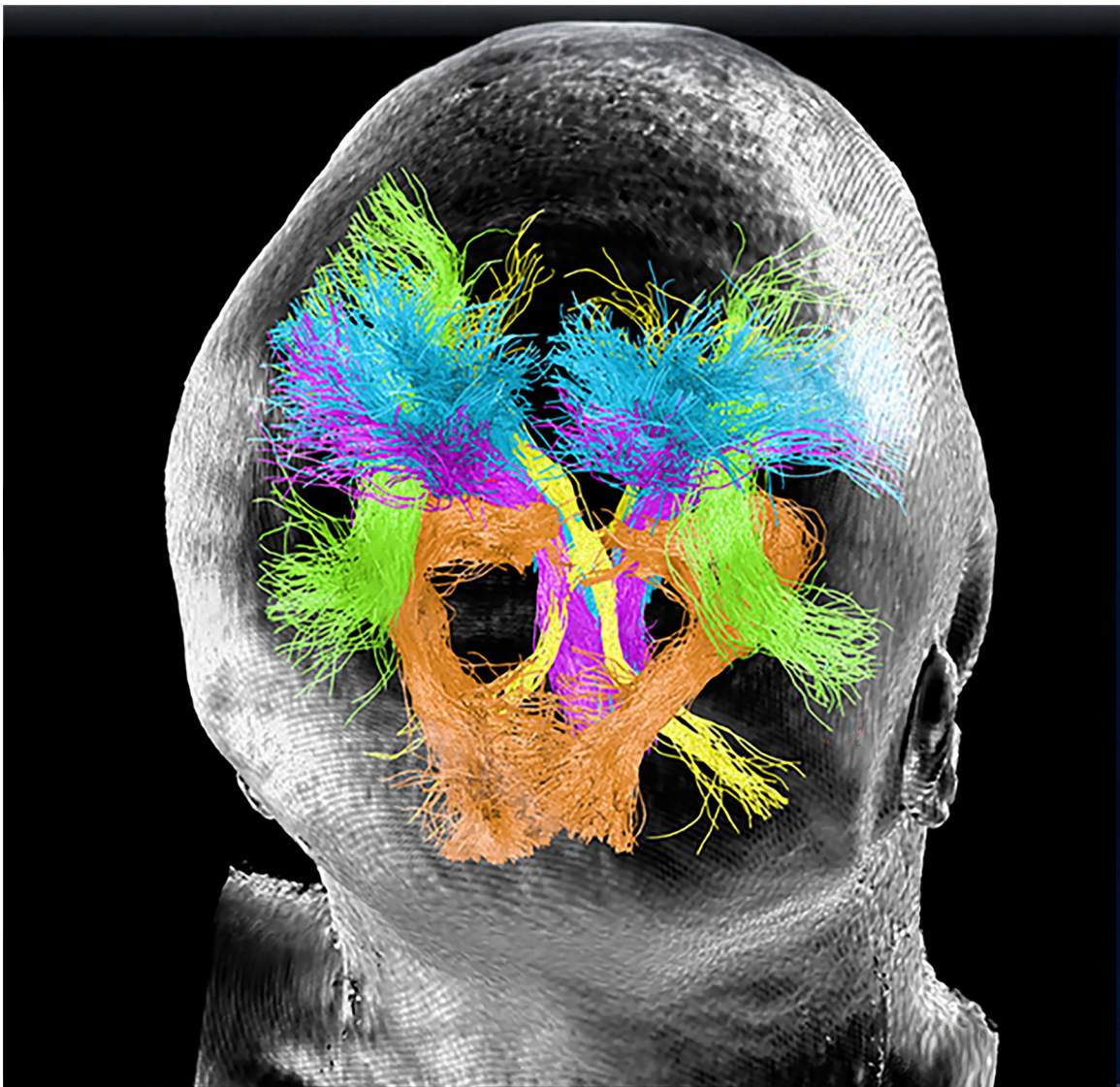




Medtronic

Stealth Tractography User Manual



Stealth AXIS™ clinical application version 1.0

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Intended use / Indications for use

The Stealth AXiS™ Surgical System, with the Stealth AXiS™ Cranial clinical application, is intended for precise positioning of surgical instruments and as an aid for locating anatomical structures in open, minimally invasive, and percutaneous neurosurgical procedures. Their use is indicated for medical conditions in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull, can be identified relative to images of the anatomy.

This can include, but is not limited to, the following cranial procedures (including stereotactic frame-based and stereotactic frame alternatives-based procedures):

- Tumor resections
- General ventricular catheter placement
- Pediatric ventricular catheter placement
- Depth electrode, lead, and probe placement
- Cranial biopsies

Contraindications

None

Device description

The Stealth AXiS™ Surgical System is a computer-assisted surgery system that is composed of a platform, clinical application, surgical instruments, and a referencing system (which includes patient and instrument trackers). The system tracks the position of instruments in relation to the surgical anatomy, known as localization, and then identifies this position on preoperative or intraoperative images of a patient. The Stealth AXiS™ Cranial clinical application helps guide surgeons during cranial procedures such as biopsies, tumor resections, and shunt placements.

Patient images can be displayed by the Cranial clinical application from a variety of perspectives (axial, sagittal, coronal, oblique) and 3-dimensional (3D) renderings of anatomical structures can also be displayed. During navigation, the system identifies the tip location and trajectory of the tracked instrument on images and models the user has selected to display. The surgeon may also create and store one or more surgical plan trajectories before surgery and simulate progression along these trajectories. During surgery, the clinical application displays how the actual instrument tip position and trajectory relate to the pre-surgical plan, helping to guide the surgeon along the planned trajectory. While the surgeon's judgment remains the ultimate authority, real-time positional information obtained through the Stealth AXiS™ Surgical System can serve to guide this judgment.

Stealth AXiS™ Core is composed of the workstation cart and the camera cart, which contain the key navigation components including the localizers and the computer with the operating system. The camera cart is used for optical procedures, and EM localizers are used for EM procedures.

Compatibility and installation

The clinical application may be used on a Stealth AXiS™ planning station in an office environment, a Stealth AXiS™ computer-assisted surgery system in an operating room or office environment, or optionally from a web browser or a connected computer if the remote planning feature has been enabled on a Stealth AXiS™ planning station.

To use the remote planning feature, the connected computer must meet the minimum system performance requirements, operating system and browser compatibility requirements, and be able to access the planning server on the network. Configuring a planning station as a planning server requires the installation of a software license by qualified Medtronic personnel and further configuration by hospital staff. See the *Stealth AXiS™ IT Guide* for server and web browser configuration instructions.

The clinical application is compatible with Stealth AXiS™ systems and Medtronic navigational instruments.

The clinical application is installed by Medtronic service personnel only. If you have any questions about your clinical application installation, contact Medtronic Technical Services.

Note: When the Stealth AXiS™ clinical application is used with other medical devices, see the user instructions for those devices for additional labeling limitations.

Navigation accuracy

Under representative worst-case configuration, the Stealth AXiS™ Surgical System with the Stealth AXiS™ Cranial clinical application has demonstrated performance in 3D positional accuracy with a mean error ≤ 2.0 mm and in trajectory angle accuracy with a mean error ≤ 2.0 degrees. The positional error is defined as the Euclidean distance from target to instrument tip at final point of insertion. The trajectory angle error is the angle between the planned surgical trajectory and trajectory of the neurosurgical instrument at final point of insertion. Depending on the surgical procedure and the particular neurosurgical instruments used, the resulting navigational accuracy may be negatively impacted.

This performance was determined using an anatomically representative phantom and utilizing a subset of system components and features that represent the worst-case combination of all potential system components in a clinically relevant workflow. The test configuration included CT images with slice spacing and thickness of 1.25 mm and T2-weighted MR images with slice spacing and thickness of 1.0 mm.

User profile

The intended user profile for the system with the Cranial clinical application is neurosurgeons, their support staff, hospital biomedical engineering support staff, medical imaging staff, and hospital IT.

The intended user profile for creating and editing tracts is users who have training in neuroanatomy and have an understanding of either MRI physics or the principles, capabilities, and limitations of tractography.

Intended patient population

The intended patient population for the Cranial clinical application is patients for whom stereotactic image-guided surgery is appropriate.

Use of the autotracts feature is intended for adult patients (22 years of age and older).

Side effects

The system and its associated applications are auxiliary surgical tools and do not independently produce side effects. When used in conjunction with navigated instruments for surgical procedures, the associated side effects are the same as those for the procedure itself.

Potential complications

The potential complications for Stealth AXiS™ Surgical System Cranial Procedures identified by Medtronic include, but are not limited to:

General surgical complications, pain, edema, hydrocephalus, insufficient treatment, infection, tissue or bone damage, burn, hearing loss, vision impairment, electric shock, cerebrospinal fluid leakage or fistula, extracranial or intracranial bleeding, brain or nervous system injury, permanent or transient neurological deficit, paralysis, seizure, death.

Clinical benefits

Navigated procedures performed using the Stealth AXiS™ Surgical System with Stealth AXiS™ Cranial clinical application potentially confer the following clinical benefits:

- Provides consistent accuracy that contributes to the achievement of surgical goals and reduces the risk of procedure-related complications for patients.
- When used with the Stealth Autoguide™ System, the system enables targeted, accurate acquisition of intracranial biopsy specimens, thereby reducing the risk of repeat procedures.
- Electromagnetic-based tracking system enables pinless cranial procedures, eliminating the need for rigid head fixation, thereby reducing patient discomfort and postoperative pain.
- Provides detailed anatomical information, including 3D models and white matter tracts visualization, supporting surgeons in developing patient-specific surgical approaches for deep-seated and/or eloquent anatomic areas, which may lessen neurological deficits.
- The use of less invasive cranial procedures may result in shorter length of stay (LOS) and reduced postoperative pain, thereby facilitating faster postoperative recovery.

Warnings

Refer to the applicable package inserts and system manual for additional instructions, warnings, and cautions.

Warning: The system hardware and software should be used only by qualified medical professionals who are trained in performing surgery and familiar with computer-assisted surgery systems.

Warning: If system navigation seems inaccurate and steps to restore accuracy are unsuccessful, abort use of the system.

Precautions

Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.

Contact information

If a serious incident occurs in relation to the use of the system, report it to Medtronic Navigation. If a serious incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

For instructions about return shipping or about transporting system components or accessories, contact Medtronic technical support.

Telephone	Technical support worldwide	+1 720 890 3160
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	General	+1 720 890 3200
	Fax	+1 720 890 3500

Mailing address	Medtronic Navigation, Inc.
	200 Medtronic Drive
	Lafayette, Colorado 80026
	USA

World Wide Web	www.medtronic.com
	www.medtronic.com/manuals

E-mail	rs.navtechsupport@medtronic.com
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Interacting with the clinical application: Touchscreen monitor and mouse

The touchscreen monitor is the primary interface for the system; therefore, these instructions include monitor interactions such as tap, double-tap, and drag. Instructions to tap an object on the screen mean to touch the monitor with a finger, a gloved finger, or a capacitive stylus and then lift the finger or stylus quickly.

You can also use a mouse to interact with the clinical application. When using a mouse, instructions to “tap” an object on the screen are equivalent to click on the object with the left mouse button.

For a list of touchscreen interactions and their equivalent mouse actions, see “Touch and mouse interactions.”

Document conventions used in this manual

This document employs the following conventions:

- Warnings are preceded by the word **Warning**. Failure to observe a warning may result in serious injury, death, or other serious adverse reactions to the patient or operator. Pay special attention to these items.
- Precautions are preceded by the word **Caution**. Failure to observe a precaution could result in injury, damaged equipment, forfeited time or effort, or the need to abort use of the system.
- References to buttons that appear on the touchscreen monitor are enclosed in square brackets. For example: Tap **[Edit...]**.
- References to buttons that are icons without text are capitalized. For example: Tap the Expand button...
- References to menu options that appear on the touchscreen monitor are printed in bold letters. For example: Select **Clear** from the list.
- References to titles of specific instructions for use documents are in italics.
- Electronic instructions for use can be accessed in two different ways: at www.medtronic.com/manuals and within the clinical application. To view the instructions for use in the clinical application, tap **[Help]**, and tap **[Go to list of instructions for use]**.

Planning options

You can create and modify surgical plans in the **Planning** task by using the workstation cart, the Planning Station, or in a remote planning session. For more information about the Remote Planning feature, see “Remote planning.” For more information about the Planning Station, see the *Stealth AXiS™ Planning Station System Manual*.

Remote planning

The Remote Planning feature allows you to plan procedures both inside and outside the operating room, provided your computer has hospital network access and the facility has set up a Planning Station server. Using a supported web browser, you can view patient data and create surgical plans in the clinical application. No patient data is stored on your computer. Plans are saved to the Planning Station server and can later be imported to a Stealth AXiS™ workstation for use in the operating room.

While a remote planning session is active, the selected patient case is locked to prevent simultaneous edits by multiple users. If another user selects the same surgeon profile, the procedure is also locked. Users are encouraged to log in with their own credentials or create their own unique profile.

Browser sessions time out after a period of inactivity, with a warning message displayed beforehand. The timeout duration is set by hospital IT staff and can last between 10 and 60 minutes.

Warning: Review patient data created using remote planning before making surgical decisions and actions.

Note: Images may take a longer time to load when using the Remote Planning feature.

Note: In a remote planning session, you cannot use a CD/DVD or USB drive to import or export images or surgeon profiles.

Minimum laptop or desktop requirements

To use remote planning, you only need to install the Google Chrome™ browser on your laptop or desktop—no additional client or software installation is required. Refer to a Windows or Apple user manual or technical support to confirm what your hardware and software specifications are.

Note: If you don't comply with the following recommendations, remote planning might not work on your laptop or desktop.

Note: If your network connection is poor, confirm that you have both a stable and fast internet connection. Consider moving your desktop or laptop to a new location.

Minimum computer screen and resolution recommendations:

- Minimum screen size: 13 inches
- The following screen resolutions have been tested:
 - 2560 x 1440
 - 1920 x 1080
 - 1440 x 810

Minimum Windows hardware and software recommendations:

- Windows 11 (64-bit) operating system
- Intel® Core™ i5* 10th generation or later processor, or equivalent AMD processor
- Memory: 16 GB RAM
- Graphics:
 - Integrated Graphics Card: Intel® Arc™* graphics card or equivalent AMD graphics card
 - Dedicated Graphics Card: NVIDIA RTX™ A1000* graphics card or equivalent AMD graphics card

Minimum Apple hardware and software recommendations

- Operating System: macOS Sonoma 14, macOS Sequoia 15
- Processor: Apple M3 or Apple M1 Pro
- Memory: 8 GB RAM
- Graphics: Apple M3 or Apple M1 Pro

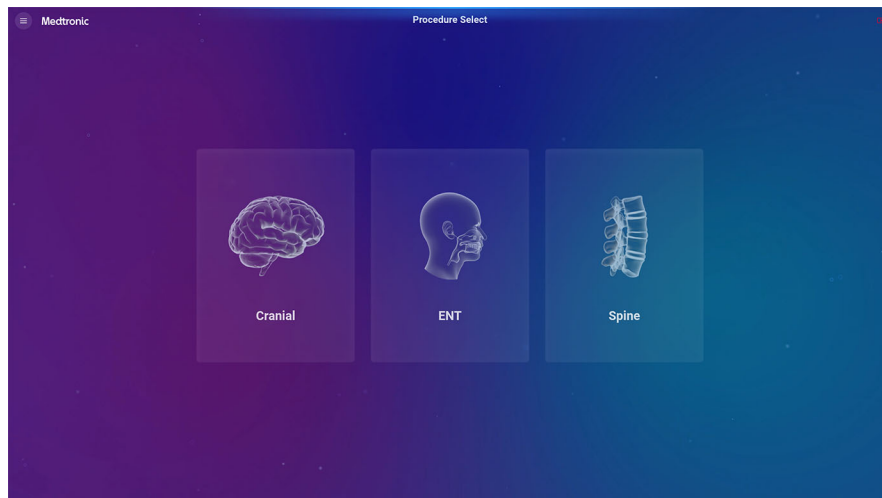
Guidance for optimizing Windows laptops or desktops

The following guidelines are recommendations for Windows laptops or desktops running Windows 11 equipped with two GPUs. Visualization performance is optimized with remote planning by using the dedicated GPU instead of the integrated GPU.

1. Confirm that the Chrome™* browser is installed on your laptop.
 - a. Press the Windows key on your keyboard, type **Graphics settings**, and then press “<Enter>.”
 - b. Click **[Add desktop app]**.
 - c. Click in the address bar, type **C:\Program Files\Google\Chrome\Application** and press “<Enter>.”
 - d. Click **chrome.exe** and then **[Add]**. Depending on your system settings, you may not see the .exe suffix. It may only be labeled “chrome”.
2. Select **Google Chrome** and click **[Options]**.
3. Select the dedicated GPU that you want to utilize.
4. Close all Chrome™* windows and restart the Chrome™* browser for the changes to take effect.

Logging in to the system from a cart

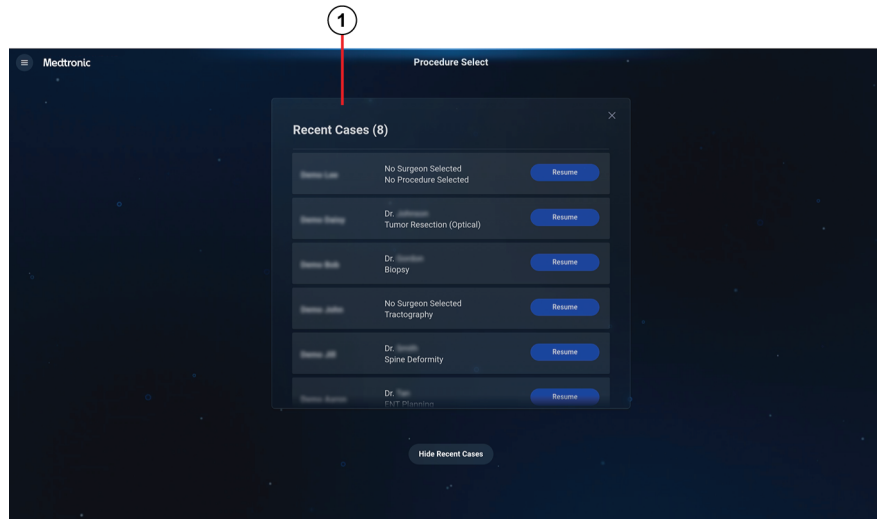
1. At the login screen, enter your username and password.
 - Tap **[Log In]**.
 - If you are unable to log in to the system, contact your hospital’s Stealth Admin or IT department to confirm login information. Medtronic technical services can provide a temporary passcode for emergency system access. Tap **[EMERGENCY AND SERVICE LOGIN]** and call Medtronic technical services for assistance.
2. To launch the clinical application, tap the desired clinical application icon or tap **[View Recent Cases]**.



Resuming recent cases

1. Tap **[View Recent Cases]** in the **Procedure Select** screen to view a list of patients imported or selected from the last 36 hours.
2. Tap **[Resume]** to select the desired patient case.

The **Images** task displays for the last surgeon and procedure used with this patient.



1 Recent Cases

Selecting a surgeon and procedure

After selecting the desired clinical application icon, the **Procedure Select** screen displays.

Follow these steps to select a surgeon and a procedure.

1. Tap **[Standard]** or select a surgeon name.
Alternatively, tap **[Add Profile]** to add a surgeon profile in the **Preferences** task. If a procedure is locked by another user, you can create a copy of the procedure.
Surgeon profiles allow you to save surgeon-specific preferences such as instruments, navigation options, and view settings. These preferences are organized by procedure. For each surgeon profile, you can also control the display of the on-screen keyboard and the colorblind aid.
2. After selecting a surgeon profile, select the surgical procedure to be performed.
Alternatively, tap **[Add Procedure]** to create or customize procedures in the **Preferences** task. See "Customizing procedures" for more details.

Images

Importing images

On a Stealth AXiS™ system, you can import patient images from multiple sources (for example, CD/DVD or USB media, other Stealth AXiS™ systems, or a hospital PACS). Images must meet certain requirements. For information on scan requirements, see the *Stealth AXiS™ Imaging Protocols for Navigated Procedures* and the *Stealth AXiS™ Tractography Imaging Protocols*.

Warning: Use care if you are using a DICOM anonymizer because the anonymizer may delete DICOM tags necessary for tractography.

You can import DICOM (Digital Imaging and Communications in Medicine) patient images in the **Images** task. Tap a tab at the top of the screen:

- Use the **Import** tab to import images from a PACS, another Stealth AXiS™ system, or CD/DVD or USB media.
- Use the **Patients On This Stealth** tab to select images that are already on the system.

Importing images from a network source (DICOM)

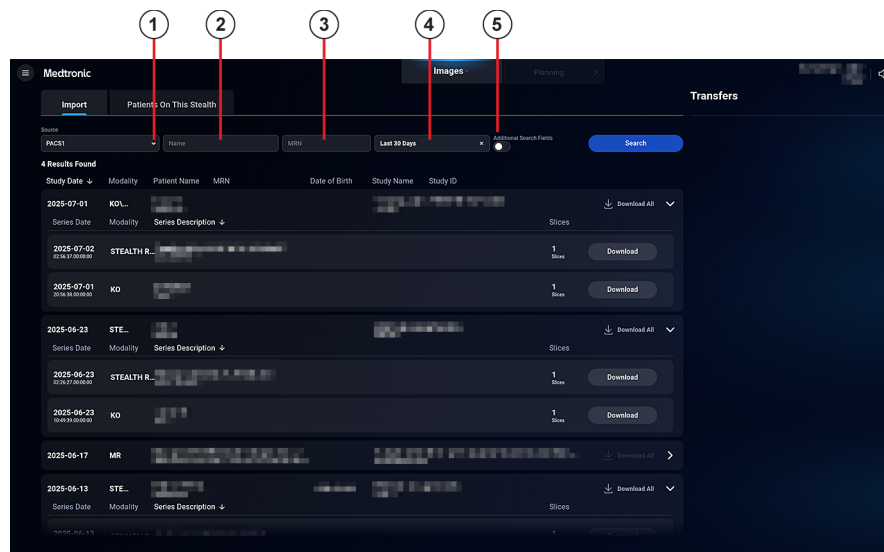
Note: If an image has been manually pushed from a network source (DICOM) to the Stealth AXIS™ system, it will automatically appear under the **Patients On This Stealth** tab.

1. Tap the **Import** tab, if it is not already selected.
2. Select a network source from the **Source** drop-down menu.
If you need to add a network source, see “DICOM Setup.”
3. Enter search criteria.
 - a. Enter information in any combination of the **Name** and **MRN** (Medical Record Number) fields. The patient name search is not case sensitive.
 - b. Tap the date filter field (**Last 30 Days** is the default filter) to display date search options (select **Last 24 Hours**, **Last 7 Days**, **Last 30 Days**, **All Time**, or a custom date range).
 - c. Turn on the **Additional Search Fields** toggle to show the **Referring Physician** field and the **All Modalities** toggle.
 - d. Turn off the **All Modalities** toggle to filter by MR, CT, or Stealth archive modalities.

Note: To see all data from the network source, leave the search fields blank.

Note: Refine search criteria for faster searches. Leaving all search fields blank may result in long search times.

4. Tap **[Search]**.



- 1 **Source** drop-down menu
 - 2 **Name** filter field
 - 3 **MRN** filter field
 - 4 Date filter field (**Last 30 Days** is the default)
 - 5 **Additional Search Fields** toggle (turns on and off additional search filters)
5. Import the images needed for the procedure. The first study in the list is expanded by default. Tap other studies in the list to display associated series.
 - To import an individual series, tap **[Download]** beside the series.
 - To import all series for the study, tap **[Download All]**.

Note: If you are importing images that were exported from a Stealth AXIS™ or StealthStation™ S8 (clinical application v2.1 or later) system, and the original DICOM images were exported also, be sure to select the Stealth archive images rather than the original DICOM images. The original DICOM images do not include any information from the Stealth AXIS™ or StealthStation™ S8 clinical application, such as merges and plans.

A list of patients displays in the **Transfers** list after a download is complete. The **Active Transfers** provides a download status for each image series.

6. After downloading all necessary images, proceed to preview the images.

Importing images from media

On a Stealth AXiS™ system, you can import patient images located on CD/DVD or USB media. This option is not available for remote planning.

1. In the **Images** task, tap the **Import** tab.
2. Insert the CD/DVD or USB media that contains DICOM images into the system.
3. If more than one media type is available, or media was already inserted before tapping the **Import** tab, select the desired media from the **Source** drop-down menu and tap **[Search]**.
4. In the **Select a directory to search** dialog, search the folders to find the desired images.

Note: The clinical application searches for DICOM images in subfolders, but manually selecting a lower level folder will result in faster search times.

5. Tap **[OK]**.
The clinical application searches for DICOM images in the selected folder.
6. Select a patient from the search results list.
7. Import the images needed for the procedure.
 - To import an individual series, tap **[Download]** beside the series.
 - To import all series for the selected study, tap **[Download All]**.
8. After downloading all necessary images, proceed to preview the images.

Viewing patient status and history

If your hospital facility is running a planning server, a lock icon next to a patient case indicates that another user has the patient case selected. If so, the system prevents you from selecting the same patient. From the **Patients On This Stealth** tab you can view the patient status on the patient card. The name of the user who has the patient currently selected displays and the patient is not selectable.

If a procedure is locked, a second user might be using the same surgeon profile to look at that procedure. For this reason, it is recommended that you create your own unique profile that is not used by others.

If you want to make a copy of a patient case while it is currently in use by someone else, you can clone the patient. See “Cloning patients” for more information.

Tap the History button to show or hide a list of entries for each previous user session. The history list includes username, time stamp, and certain actions such as plans created. This may be helpful to detect unexpected edits to the plan by someone else. The History button is disabled if the patient is currently selected by another user on the system.

Tractography

Tractography is based on the movement of water in the brain. Water naturally diffuses randomly throughout the brain due to Brownian motion unless there are structures like fiber tracts that restrict the diffusion of water to be along the axis of those fibers. The premise of white matter tractography is to infer where the white-matter fiber tracts are by finding areas of the brain where water is diffusing in a more linear manner.

Limitations of tractography include the following factors:

- Because tractography is based on the diffusion of water in the brain, any condition that alters the flow of water may alter the fiber tract results. For example, the presence of edema around a tumor, which is like a pool of water, can result in an incomplete tract.
- Because fiber tracts cross one another throughout the brain, tractography may result in a tract that crosses from one fiber tract to another or an incomplete tract. To refine fiber tracts, see “Editing custom tracts.” You can import DTI tracts from earlier versions of Stealth tractography, but only display properties are available.

Warning: Performing tractography with poor quality scans resulting from high noise or motion artifacts, or in the presence of known pathologies or edema, may result in generated fiber bundles that do not represent anatomically correct white matter fiber tracts. Always corroborate generated tracts with all available patient information and your anatomical knowledge.

A Stealth Tractography license is required to create and edit tracts. Without a license, you can still display tracts, change display properties, view tract creation parameters and verify autotracts.

Automatic tractography

The automatic tractography feature (autotracts) automatically generates fiber tracts selected during diffusion processing.

If you are using a system without a Stealth Tractography license, autotracts cannot be generated. However, you can verify and display autotracts when you import them from a system with the license.

AI transparency statement for autotracts

The autotracts feature automatically generates white matter tracts of the brain from preoperative diffusion scans using locked machine learning and deep learning algorithms. The algorithm has been trained and tested using data mainly from GE and Siemens scanners. Use of diffusion scans from other scanner manufacturers may yield tracts with fewer fibers or noisier results.

With the autotract result, the user can crop fibers they believe to be extraneous and adjust the fiber probability slider. Prior to use in planning or navigation tasks, the user must review and verify each automatically generated tract. Users also have the option to manually create custom tracts.

Previewing and processing diffusion images

Before proceeding to the next task, preview any images that you plan to use for navigation and process the diffusion series that you plan to use for creating tracts.

Previewing images

When your patient's name displays in the list of images:

1. Tap the patient's name. The clinical application displays all of the series associated with the patient in the **Patient Images** list.

Note: If exams are not under the same patient, see "Managing patients and images" for instructions on how to combine patients or move images.

2. Tap on the series that you plan to use for navigation. The clinical application displays preview images and image details. Options to browse and view additional slices include:

- Tap and drag the slice indicator on the vertical scroll bar or use the mouse wheel.
- Tap above or below the slice indicator to move one slice in the indicated direction.
- Tap the arrow at the top or bottom of the scroll bar to move one slice in the indicated direction.

3. In **Exam Details**, tap the Expand button to view more image details.

4. Preview the images.

- a. Make sure that the anatomy of interest is captured in the images.
- b. Make sure that the images do not contain geometric distortion or intensity distortion (intense brightness in parts of the image compared to the rest of the image). If images contain distortion, see "Image import problems."
- c. Check the slice spacing and slice thickness. Series with slice spacing equal to slice thickness are recommended.
- d. To change the view orientation of a selected procedure, see "Customizing procedures." If you have not selected a procedure, you can change the view and patient orientation while in this task.

5. If the image orientation is incorrect in the scan, you can correct it:

Warning: Only correct the image orientation if it is incorrect in the scan. Verify that all aspects of the patient images are correct, including patient data and image orientation. Make sure that the image orientation matches the patient's anatomy. Incorrect image orientation can result in inaccurate navigation and loss of data.

- a. To rotate the image, tap **[Correct]** and tap the desired icon in the **Rotate** tab. Tap **[Confirm]**.
- b. To reorient the image, tap **[Correct]** and tap the **Reorient** tab. Then tap the desired orientation button. Tap **[Confirm]**.

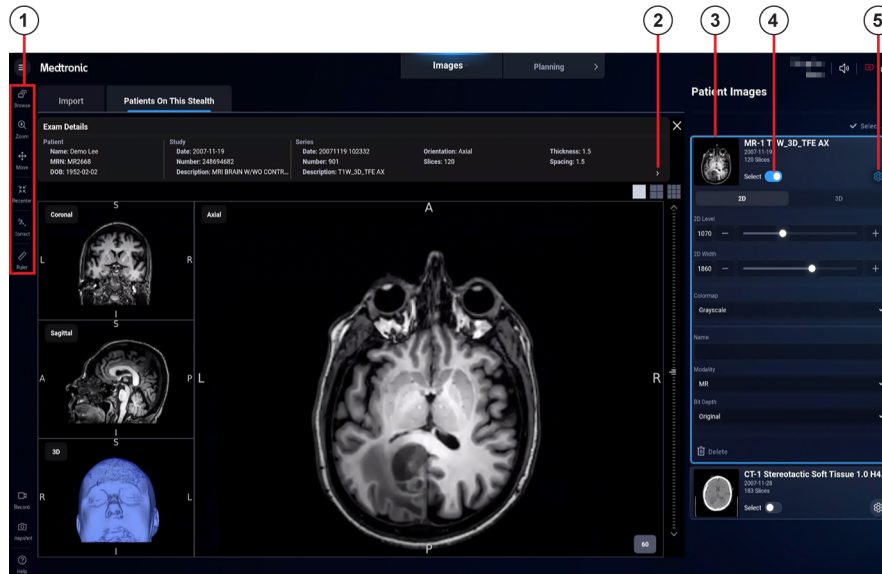
6. If you need to adjust the level or width, tap the Settings button on the series in the **Patient Images** list. Then tap the **2D** or **3D** tab and use the **Level** and **Width** sliders.

For additional information about the buttons on this screen, see "Buttons in Images, Tractography, or Planning."

7. If you plan to merge multiple series from the same patient:

- a. Tap each series in the **Patient Images** list to preview the images and check the image quality.
- b. Turn on the **Select** toggle in an image series to add the images to the merge set.

8. After confirming that everything is correct in the patient's images, tap the next desired task at the top of the screen.



- 1 Tools for modifying images
- 2 The Expand button displays additional exam details
- 3 Patient exams
- 4 The **Select** toggle is only available when the Cranial StealthMerge™ license is installed
- 5 The Settings button shows image display options such as level and width adjustments

Processing diffusion images

When your patient's name displays in the list of imported images:

1. Click the patient's name. The clinical application displays all the series associated with the patient in the **Patient Images** list.

If exams are not under the same patient, see “Managing patients and images” for instructions on how to combine patients or move exams.

2. Click the diffusion series that you plan to use for creating fiber tracts. The clinical application displays preview images and image details.

Note: If multiple diffusion series are available for the patient, select only one diffusion series to use for creating tracts. If the scanner software created a version of the diffusion series that includes motion correction, consider using that version.

For more image details, tap the Expand button.

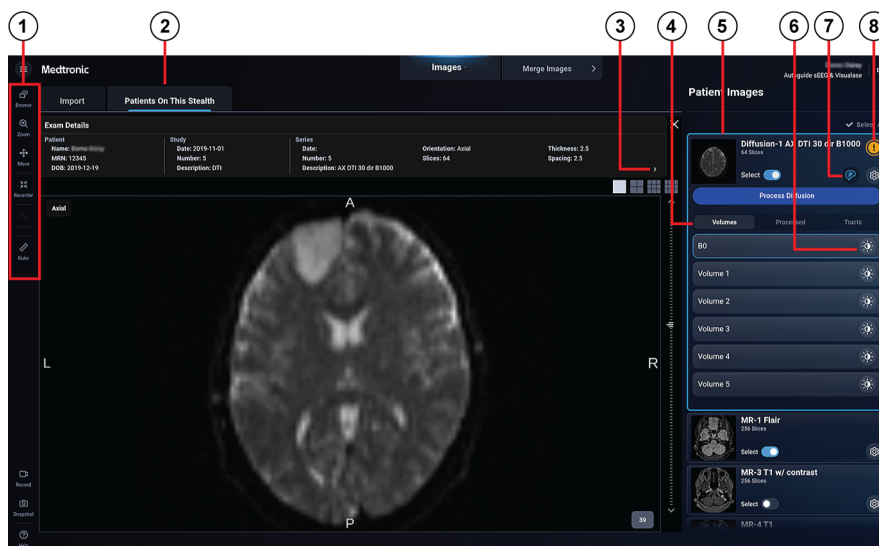
To browse additional slices, drag the scroll bar. Optionally, click one of the buttons above the view to change the number of images on the screen.

3. Preview the images.
 - a. Make sure that the anatomy of interest is captured in the images.
 - b. Make sure that the images do not contain geometric distortion or intensity distortion (intense brightness in parts of the image compared to the rest of the image). If images contain distortion, see “Image import problems.”
 - c. Check the slice spacing and slice thickness. Series with equal slice spacing and slice thickness are recommended.

Warning: Verify that all aspects of the patient images are correct, including patient data and image orientation. Make sure that the image orientation matches the patient's anatomy.

If you need to adjust the level or width, tap the brightness button on the series in the **Patient Images** list.

For additional information about the buttons on this screen, see “Buttons in Images, Tractography, or Planning.”



- 1 Tools for modifying images
- 2 Patient data
- 3 Expand button
- 4 **Volumes** tab
- 5 Displays image information
- 6 Brightness button: displays options to adjust the level and width of an image
- 7 Diffusion button
- 8 Warning or error icon: expand the card to click the **[Warning]** for more details.

4. If the list of gradient volumes is not visible, click the Diffusion button on the diffusion series and then click the **Volumes** tab. The list of diffusion gradient volumes displays.

- The clinical application only requires one B0 volume. If multiple B0 volumes are in the dataset, the one being used by the clinical application is shown with white text. The other B0 volumes are shown with gray text. You can click on the gray text to view the B0 volume, but you cannot change which B0 volume the clinical application uses.
- To quickly preview the different gradient images within a volume, scroll through them with the mouse wheel or swipe up or down with your finger. The gradient images display in the image view.

Note: The B0 can be displayed in axial, coronal, and sagittal orientations, but the other gradient volumes can only be displayed in the orientation in which they were acquired.

5. Follow these steps to review the gradient volumes and check for patient motion between the volumes:

Warning: Make sure the gradient volumes are correct before creating tracts. If the gradient volumes are incorrect, the subsequent tractography will not be accurate.

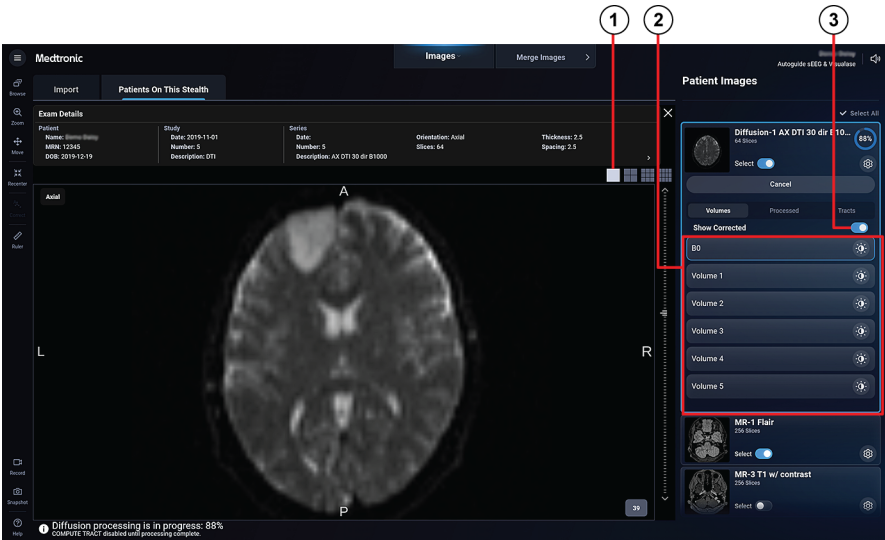
- a. Make sure the view displays one slice of the gradient volume at a time.
- b. Use the mouse wheel or the <PgUp> and <PgDn> keys to scroll through the list of gradient volumes and look for movement of the brain between volumes.

- c. Look for volumes with incomplete information or excessive noise.

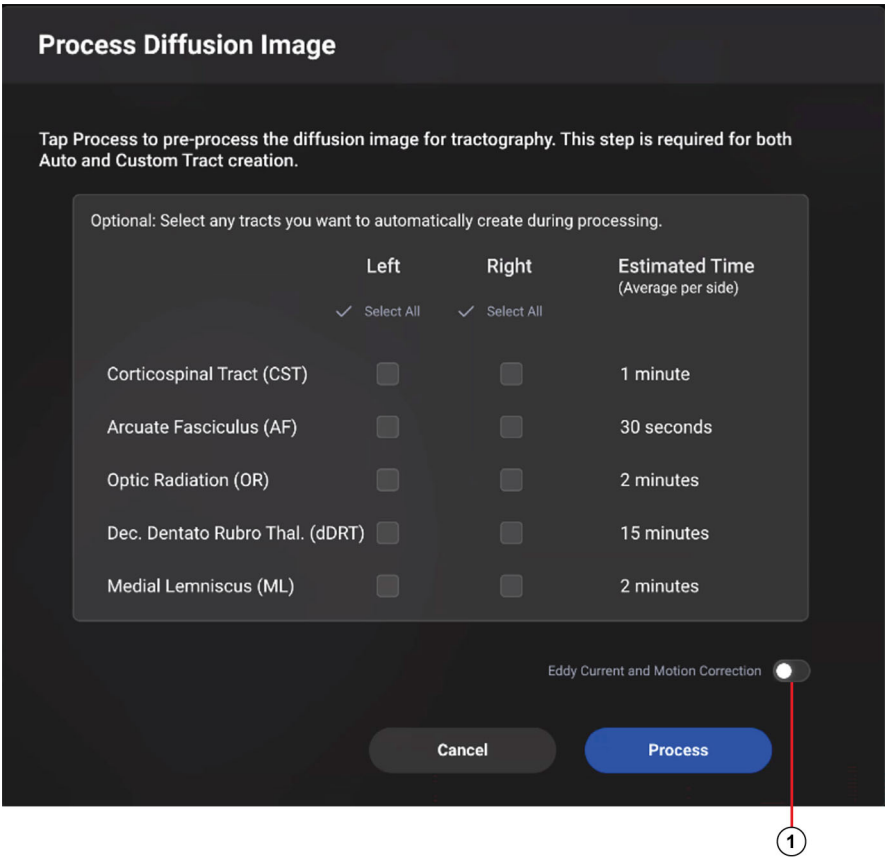
If you see gradient volumes with incomplete information or excessive noise during your review, carefully assess the DEC map and any fiber tracts after diffusion processing. If the DEC map or fiber tracts are not adequate for your purposes, rescan the patient to improve the diffusion scan quality.

- d. Look for distortion in the gradient volumes.

If the area of interest appears too distorted, consult with Radiology. See also *Stealth AXiS™ Tractography Imaging Protocols*.



- 1 Display one slice of the gradient volume at a time
- 2 Diffusion gradient volumes
- 3 Show Corrected



- 1 Eddy Current and Motion Correction toggle

6. To start processing the diffusion series click [Process Diffusion]. On the Process Diffusion Image dialog:

- a. If you need to correct for eddy current distortions or for patient motion, turn on the **Eddy Current and Motion Correction** toggle.

Note: Eddy Current and Motion Correction is off by default. **Eddy Current and Motion Correction** attempts to correct the distortion caused by eddy currents and motion artifacts. Using **Eddy Current and Motion Correction** increases the time required to process the diffusion series.

- b. Select the desired tracts you would like to have automatically created. Autotracts already computed for the patient will show as checked.
- c. Click **[Process]**.

A progress bar displays while the series is being processed. When processing is complete, the processed data (for example, DEC Map, ADC Map, FA Map, or Brain Mask) display on the **Processed** tab. Autotracts display on the **Tracts** tab.

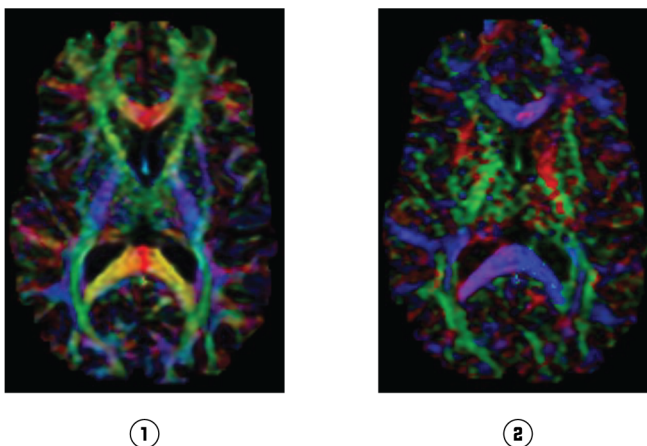
7. If you used **Eddy Current and Motion Correction**, review the gradient volumes for motion again. If the correction is not adequate, rescan the patient.

Note: When **Show Corrected** is switched on, the image view displays the eddy current and motion corrected scan direction. You can switch **Show Corrected** off to display the native scan direction.

8. On the **Processed** tab, click a processed map to view it in the image view. Review the **DEC Map** colorization.

Warning: Review the DEC Map colorization before creating tracts. Use your knowledge of white matter anatomy to determine whether the DEC colorization matches your expectations. If the colorization is not correct, do not perform tractography.

If the DEC Map colorization is not correct, consult with Radiology to make sure that the patient was scanned according to the *Stealth AXiS™ Tractography Imaging Protocol*. If the imaging protocols were not followed, rescan the patient.



- 1** Correct DEC coloring
- Green: Anterior-Posterior
 - Red: Right-Left
 - Blue: Superior-Inferior

- 2** Incorrect DEC coloring

9. Follow these steps to review the Brain Mask:

- a. On the **Processed** tab, click **Computed Brain Mask**.

The Brain Mask displays in the image views.

- b. Make sure the Brain Mask follows the envelope of the brain and there are no holes in the Brain Mask near the area of interest. If there are holes in the Brain Mask, the fiber tracts will not be built in the area with holes.

Note: Review the Brain Mask for correctness before creating tracts. The Brain Mask is not editable. If there are erroneous areas of the Brain Mask near the area of interest, consider whether to proceed with tractography using this scan.

Small, extraneous areas outside of the brain (for example, the sinuses or skull) will not affect the completeness of the generated fiber tracts.

10. On the **Tract** tab, check that the expected autotracts have been generated. You can review and verify autotracts in the **Tractography** task after previewing other images and merging image series.

Using image merge

The image merge feature is only available when the Cranial StealthMerge™ license is installed. Image merge enables you to merge multiple images from the same patient.

See “Merging a diffusion series to a diffusion reference series” for additional considerations when working with these types of images.

Warning: Merging patient images manually may impact the overall system accuracy. Always visually verify the accuracy of each merge before using it for planning and navigation. Failure to confirm accuracy can result in inaccurate navigation.

Warning: In some cases, dissimilar image modalities may not accurately merge with one another. Images with similar anatomical details are likely to merge accurately, whereas images with significantly different details may not merge accurately in every case. For best results:

- When merging more than two images, use the highest contrast image as the reference image. For example, use a pre-operative, high-resolution, diagnostic-quality MR-T1 or CT as the reference image.
- When merging diffusion images, use a pre-operative, high-resolution, diagnostic-quality MR as the diffusion reference image.
- When merging O-arm™ system images, you can use any of its imaging modes.
- Make sure that a significant portion of the skull base is visible in all images so that there are enough anatomical details to complete an accurate merge.

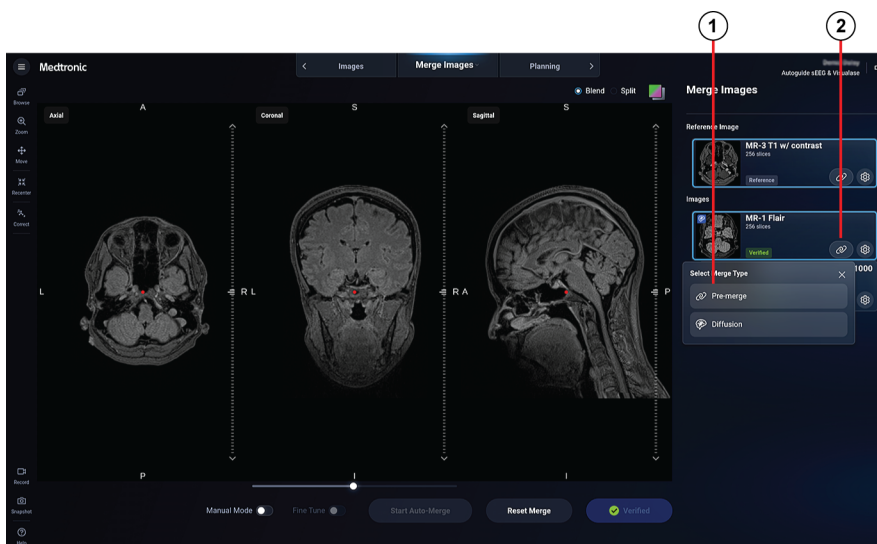
Warning: When merging diffusion series, use a pre-operative, high-resolution, diagnostic-quality MR as the reference series for best results. A diffusion series may not merge accurately with a CT image or a 3D intra-operative image (such as an O-arm™ system image). If you merge a diffusion series with a CT image or a 3D intra-operative image, thoroughly check the accuracy of the merge.

In the **Images** task, turn on the **Select** toggle in an image series to add the images to the merge set. To remove an image series from the merge set, turn off the **Select** toggle in the image series. You can merge images in the **Merge Images** task. If you load or acquire more images later in the procedure, you can return to the **Merge Images** task to merge additional images.

Merging a non-anatomical image series to an anatomical image series

If an image series contains non-anatomical images (such as a PET or fMRI), follow these steps to associate the series with an anatomical image series with the same frame of reference. Note that this presumes the patient did not move between the acquisitions of the two series.

1. Tap the image card for the image that will be the reference image. This pre-merge reference series is not required to be the navigation reference series.
2. Tap the Link button and select **[Pre-merge]**.
3. Tap the image card for the image you want to pre-merge to the reference image.
4. Tap **[Pre-merge]**.
5. Verify that the match between them looks correct.
6. To unlink pre-merged images, tap the Settings button on the desired image and follow one of these steps:
 - Tap **[Unlink Pre-Merge]** on a pre-merge image to break the link to the reference image. The pre-merge image moves back into the **Images** list.
 - Tap **[Unlink All Pre-merged Images]** on the pre-merge reference image to break the link to all pre-merged images. The pre-merge reference image and all other images move back into the **Images** list.



1 [Pre-Merge]

2 Link button

Merging multiple anatomical image series

To merge multiple anatomical image series from the same patient:

1. Tap **[Merge Images]**.

The **Merge Images** screen appears.

The clinical application automatically selects a reference image and starts merging the first image in the **Series** list to the reference image. To change the reference image, tap the Settings button in the series you want as the reference and then tap **[Set As Reference]**.

Note: An fMRI series cannot be used as the reference exam.

Note: If you are merging an O-arm™ system image with a pre-operative image, select the pre-operative image as the reference image.

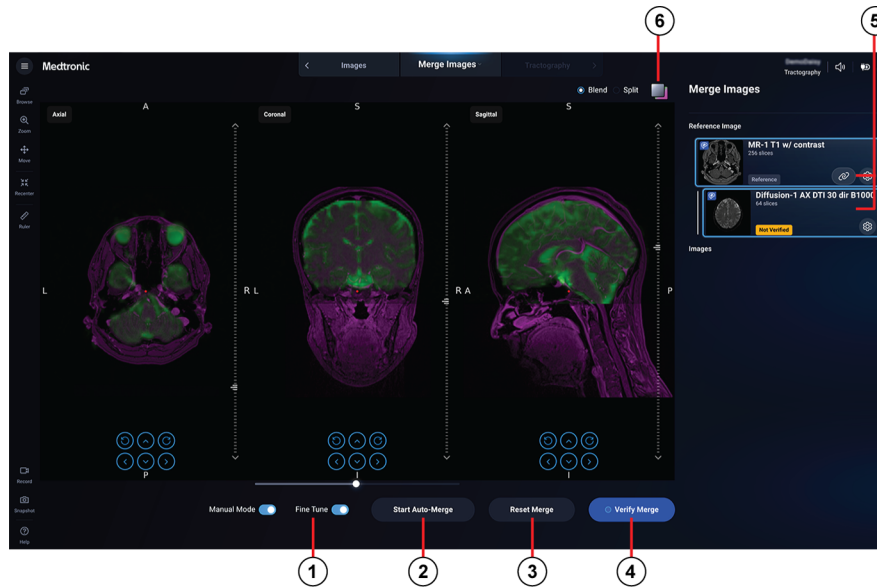
Warning: If you change the reference image after you merge images, the clinical application deletes any merges of images that had been automatically merged (such as CT and MR images). Then you must re-merge images to the new reference. Changing the reference image does not delete inherently aligned images (that is, images aligned using pre-merge), or the merge of a diffusion series to a diffusion reference exam.

Note: If the diffusion reference exam is different from the reference exam for the rest of the merge set, changing the reference exam for the merge set does not break the association between the diffusion exam and the diffusion reference exam. Changing the reference exam for the merge set also does not break the association between the diffusion reference exam and the tracts.

2. Check the accuracy of each merged exam. To check merge accuracy, tap **Blend** or **Split** and scroll through the images.
 - If the accuracy of the merge is sufficient for the procedure, tap **[Verify Merge]**.
 - If the accuracy of the merge is not sufficient, but the automatic merge (Auto-Merge) appears to be converging to the correct alignment, you may try tapping **[Start Auto-Merge]** again to allow Auto-Merge to continue registering.
 - If you want to fine tune the merge, tap **Manual Mode** and follow these steps:
 - a. Tap **Fine Tune**. Directional arrows and rotate buttons appear in the Axial, Coronal, and Sagittal views.
 - b. Use the directional arrows to move the image 0.1 mm at a time.
 - c. Use the rotate buttons to rotate the image 0.1 degree at a time.
 - d. When the image to merge is aligned with the reference image, tap **[Verify Merge]**.
 - If the merge is not aligned correctly, tap **[Reset Merge]**. Turn on **Manual Mode**.
 - a. Drag the image to merge into approximate alignment with the reference image.
 - b. When the image is aligned with the reference image, tap **[Start Auto-Merge]**.

- After you tap **[Verify Merge]**, the next merged image displays for your verification.

When you merge images, the merged set is always used in the **Tractography**, **Planning**, **Registration**, and **Navigation** tasks. See “Adjusting views for images and models” if you want to change the merged display in these tasks.



- Fine Tune**
- [Start Auto-Merge]**
- [Reset Merge]**
- [Verify Merge]**
- Selected images for merge with Link button
- Color button

Merging diffusion series images

To keep the relationship between the tracts and the images the tracts were built from (B0), the diffusion series is allowed to merge to a different reference series known as the diffusion reference. The goal is to maintain the relationship between the diffusion exam and the diffusion reference exam to which it is merged.

Merging a diffusion series to a diffusion reference series

For a diffusion series, the clinical application automatically attempts to select a diffusion reference series with preference for a series in the same frame of reference (that is, acquired in the same MR imaging session) and with the smallest slice spacing. If the clinical application does not automatically select a diffusion reference series, follow these steps:

- Tap the image card to select the reference image.
- Tap the Link button and select **Diffusion**.

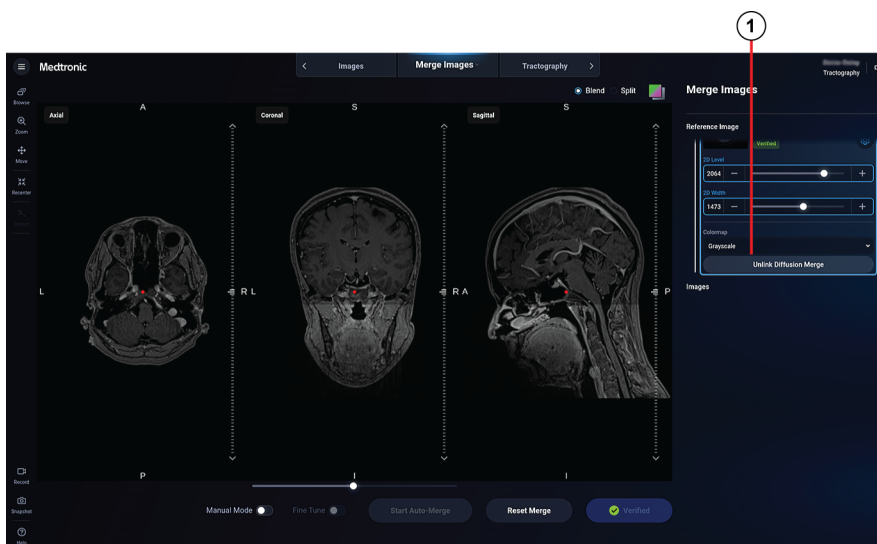
Warning: When merging diffusion series, use a pre-operative, high-resolution, diagnostic-quality MR as the reference series for best results. A diffusion series may not merge accurately with a CT image or a 3D intra-operative image (such as an O-arm™ system image). If you merge a diffusion series with a CT image or a 3D intra-operative image, thoroughly check the accuracy of the merge.

Note: The merge set can only include one diffusion image series.

If you need to break the association between a diffusion series and a diffusion reference series, use one of these options:

- Tap the diffusion merge series and tap **[Unlink Diffusion Merge]**.
- Tap the diffusion merge reference image and tap **[Unlink Diffusion Merged Image]**.

The diffusion image series can merge to a different reference image than the rest of the exams in your merge set. This allows the diffusion image series to merge to a diffusion reference image (for example, an MR). Then the diffusion reference image and the rest of the images in the merge set can be merged to an MR or a CT (merge reference image).



1 [Unlink Diffusion Merge]

Changing the merge reference image after a merge

If you acquire a new intra-operative O-arm™ system image it does not need to be the new merge reference exam. If you do change from a pre-operative T1 or T2 to the intra-operative CT-O-arm as the reference exam, note that some data is transferred to the new merge reference exam while other data is not. If you change the top (primary) reference exam, the auto-merged exams are broken but the merged diffusion exams and pre-merged exams are preserved.

Plans can shift based on the merge accuracy. If you planned on a merged set of exams, and then you change the navigation reference image for your merged set, make sure the accuracy of the merge to the new navigation reference image is sufficient for the procedures before you tap **[Verify Merge]**. When you verify the merge, any existing plans are automatically transferred to the new merged set.

Warning: Verify that a plan's target point, entry point, and trajectory are clinically valid before using it for a procedure. Traverse the surgical plan to check the entry, target, and trajectory after any edit. If you change the merge reference image after the plan is created, review your surgical plan to avoid inaccurate navigation. Some existing plan data is remerged to the new reference image and may have shifted.

Tractography task

Editing and reviewing generated autotracts

Autotracts are created during diffusion image processing in the **Images** task. In the **Tractography** task, you can use a variety of tools to edit and review generated autotracts.

Warning: Review automatically segmented tracts for correctness and only verify tracts that are useful for clinical purposes. Working with a clinically poor tract may result in a clinically poor plan.

Warning: Corroborate generated tracts with your anatomical knowledge and all available patient information. Working with a clinically poor tract may result in a clinically poor plan.

To edit an autotract:

1. To change the probability selection, use the **Probability: More/Less Fibers** slider on the autotract card. Each fiber has a probability of being a member of the tract. Moving the slider changes which fibers are displayed based on their probability.
2. To crop, see "Cropping fibers in autotracts and custom tracts."
3. Click **[Verify Tract]**. Verified autotracts display in the **Planning** and **Navigation** tasks.
4. Click **[Reset]** to revert the tract back to the original probability and undo any cropping.

An autotract reverts to the unverified state if the probability is modified, diffusion processing is updated, or the autotract is reset. An unverified label appears on the autotract card.

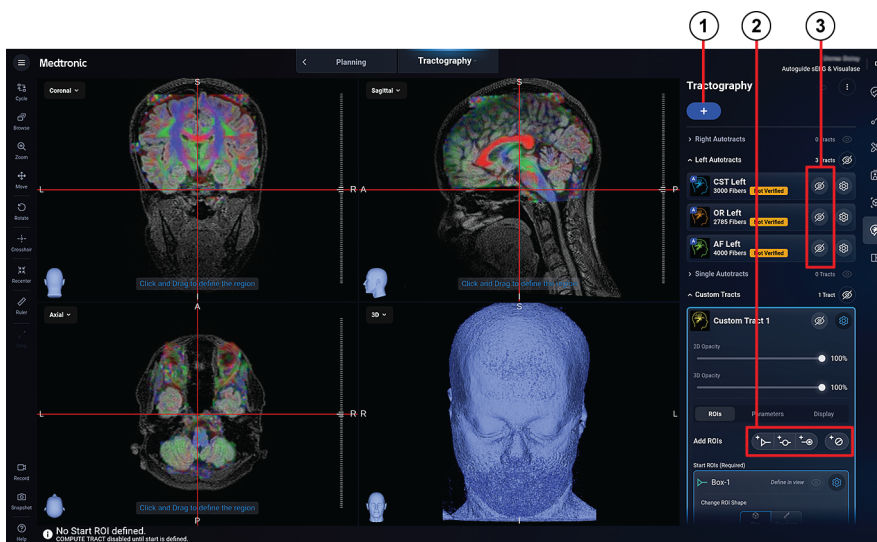
Building custom tracts

To build a custom fiber tract, you must define at least one start region of interest (ROI). Mid, end, and exclude ROIs are optional, and you can create multiple start, mid, end, and exclude ROIs. If you define mid and end ROIs, the fibers in a tract must go through a start, any mid, and any end. Fibers that do not follow the progression through start, mid, and then end ROIs are automatically excluded. Tracts do not stop at end ROIs. Tracts go through end ROIs.

Warning: Corroborate generated tracts with your anatomical knowledge and all available patient information. Working with a clinically poor tract may result in a clinically poor plan.

Follow these steps to build a fiber tract:

1. In the **Tractography** drawer, click **[+]** to create a new custom tract.
When you are creating ROIs on small anatomical regions of the brain, use **[Zoom]** to magnify the structure of interest as much as possible. Click **[Zoom]** and drag up on an image to increase magnification (or right mouse click and drag up or down to adjust magnification).
2. On the ROIs tab, click one of the four **Add ROIs** buttons: Start ROI, Mid ROI, End ROI, or Exclude ROI.
3. On the ROI card that is created, select the ROI shape that you want to define: **[Box]** or **[Freeform]**.
If you are using a box or freeform shape, click and drag or paint in one of the 2D image views to define the ROI.
You can define the ROI in any of the 2D image views. Click in the middle of the ROI to see it in all three 2D views. Then, you can redefine the ROI in any or all of the 2D image views.
4. To redefine the region of a box ROI, take one or more of the following actions in a 2D image view:
Note: When you click on or hover over part of an ROI, an icon appears in the upper-right corner of the image view to indicate whether you can move, resize, or rotate the ROI.
 - a. Click the edge of the ROI and drag the edge to increase or decrease the ROI size.
 - b. Click the corner of the ROI and drag to rotate the ROI.
 - c. Click inside the ROI and drag to move the ROI.**Note:** If two ROIs are overlapping, edits affect the current ROI. To make a different ROI current, click the name of the ROI in the list of ROIs. You can also click on a ROI's border to make it become the current ROI.
5. To redefine the region of a freeform ROI, select the ROI card that you want to change.
 - a. To add to the ROI, click **[Paint]** and then start painting to enlarge the existing freeform shape, or draw another freeform shape to make disjointed regions part of one ROI.
 - b. To reduce the ROI, click **[Erase]** and then click and drag on the freeform shape in a 2D image view.**Note:** The zoom level affects how much of the area is painted or erased. Zoom out to cover more area and zoom in for more fine-tuned edits. The resolution of the freeform region is the voxel size of the diffusion series.
6. To add a mid or end ROI, click **Mid ROIs** or **End ROIs** at the top of the ROIs tab.
If you want to add an exclude ROI, see "Editing custom tracts" for more information.
7. Select the ROI shape that you want to define for the **Mid ROI** or **End ROI**: **[Box]** or **[Freeform]**.
8. Click and drag in one of the 2D image views to define the ROI.
You can define the ROI using any of the 2D image views. Then, you can redefine the ROI in any or all of the 2D image views.
9. Click **[Compute Tract]**.
When the computation is complete, the clinical application displays the tract in the 2D and 3D image views.
10. To add another tract, click **[+]** in the upper right corner of the **Tractography** drawer.
When you add a new tract, the clinical application automatically hides all other tracts in the 2D and 3D image views. To make another tract visible, click the Show/Hide button on its tract card.
Note: If you are planning to import the fiber tracts into another Medtronic clinical application that is compatible with the Stealth Tractography feature, you may need to iterate to try different numbers of fiber lines to determine the optimal number that can be rendered simultaneously in that clinical application. The number of fiber lines displays under the tract name in the **Tractography** drawer.



- 1 **[+]** to add a tract
- 2 Click one of the **Add ROIs** buttons to add start, mid, or end ROIs
- 3 Show/Hide buttons

Editing custom tracts

You can edit tracts by adding or deleting ROIs, redefining the region of ROIs, changing the tract parameters or cropping fibers. If you make edits to a tract that requires the clinical application to recompute it, **[Compute Tract]** turns blue (active) and a message prompts you to run the computation to apply your changes. The following changes require the clinical application to recompute the tract:

- Adding or deleting an ROI
- Redefining the region of an ROI
- Changing the parameters

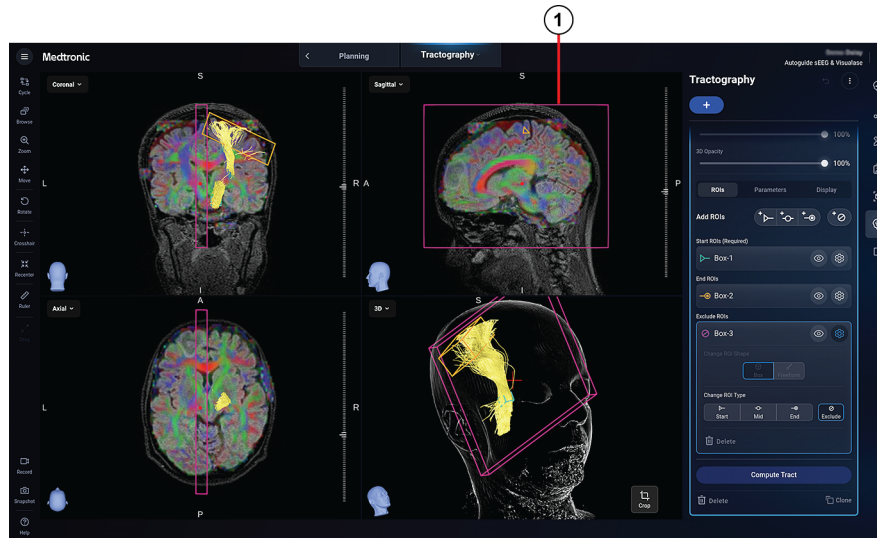
Warning: Corroborate generated tracts with your anatomical knowledge and all available patient information. Working with a clinically poor tract may result in a clinically poor plan.

Note: Consider changing the 3D view to Glass mode before adding exclude ROIs or cropping fibers. Glass mode allows you to see through outer layers of 3D models, such as skin and bone, to increase the visibility of other models, such as tumor models. Glass mode also allows you to click inside the brain (perhaps on extraneous fibers) in the 3D view to move the crosshairs to that location in the 2D views.

If you want to eliminate extraneous fibers from a tract, you can add exclude ROIs or crop fibers. Exclude ROIs remain when you recompute the tract. The results of cropping fibers are not preserved when you recompute a tract.

See “Cropping fibers in autotracts and custom tracts” if you want to crop fibers. If you want to add an exclusion region:

1. Click **[Exclude]** on the ROIs tab of the tracts card.
2. Select the ROI shape.
3. Click and drag in a 2D image view.



1 Exclude ROI

Editing ROIs

You can change clone an existing tract to experiment with edits, change an ROI type, redefine the region of a box or freeform ROI, or delete an ROI.

Note: After an ROI has been defined, you cannot change the ROI shape. For example, you cannot change a defined box ROI to a freeform ROI.

- To clone a tract, click **Clone**. A new tract card displays in the drawer.
- To reassign an ROI type (for example, to change a **Mid** to an **End**), on the **ROIs** tab of the tract card, select the ROI that you want to change, click the Settings button, and click the new ROI type.
- To redefine the region of a box ROI, take one or more of the following actions in a 2D image view:

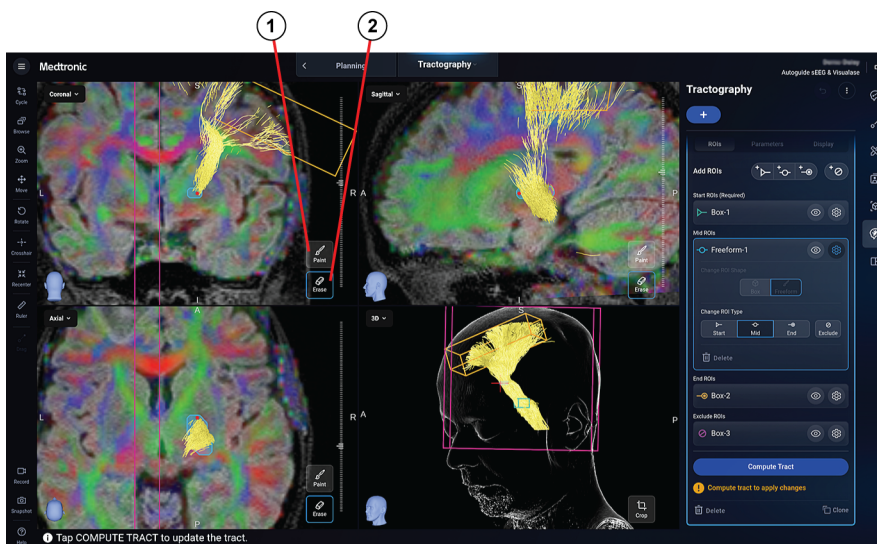
Note: When you click on or hover over part of an ROI, a hint appears in the upper-right corner of the image view to indicate whether you can move, drag the edge, or rotate the ROI.

- Click the edge of the ROI and drag the edge to increase or decrease the ROI size.
- Click the corner of the ROI and drag to rotate the ROI.
- Click inside the ROI and drag to move the ROI.

Note: If two ROIs are overlapping, edits affect the current ROI. To make a different ROI current, click the name of the ROI in the list of ROIs. You can also click on a ROI's border to make it become the current ROI.

- To redefine the region of a freeform ROI, on the **ROIs** tab of the tract card, select the ROI that you want to change.
 - To add to the ROI, click **[Paint]** and then start painting to enlarge the existing freeform shape, or draw another freeform shape to make disjointed regions part of one ROI.
 - To reduce the ROI, click **[Erase]** and then click and drag on the freeform shape in a 2D image view.

Note: The zoom level affects how much of the area is painted or erased. Zoom out to cover more area and zoom in for more fine-tuned edits. The resolution of the freeform region is the voxel size of the diffusion series.



1 [Paint]

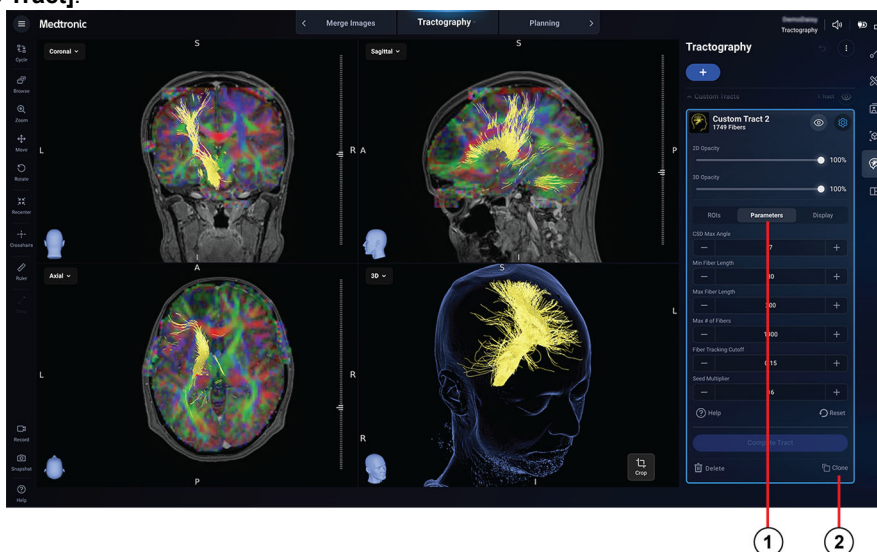
2 [Erase]

- To delete an ROI, on the **ROIs** tab of the tract card, select the ROI that you want to delete, click the Settings button, and click **[Delete]**. Click **[Delete]** on the dialog to confirm you want to remove the ROI.

Changing custom tract parameters

To change a tract's parameters, first create a copy of the tract. This allows you to make separate edits for comparison.

- Click **[Clone]** on the tract card to create a copy of the tract.
- Click the **Parameters** tab of the tract card to display the parameters.
- For each parameter you want to change, click **[+]** or **[-]** or enter in a value. Click **[Reset]** to return all parameter values to their default values.
- Click **[Compute Tract]**.



1 Parameters tab

2 [Clone] tab

See "How the Enhanced CSD tract fidelity works" for more information about the parameters.

Cropping fibers in autotracts and custom tracts

Note: The results from cropping fibers are not preserved when you recompute a tract. To preserve fiber cropping when recomputing, use exclude ROIs.

Note: The crop only affects the current tract. The clinical application finds the whole fiber that is partially overlapped by the crop box and then deletes the whole fiber (that is, it does not cut off just the part of the fiber inside the crop box).

If you want to crop an autotract or custom tract, follow these steps:

1. Rotate the 3D view to make sure the desired fibers to be deleted are visible and isolated. The crop will extend through the volume.
2. Click **[Crop]** in the 3D image view and then click and drag in the 3D image view. After you release, the clinical application performs the crop. Crop mode is automatically turned off so that you can then rotate the 3D view to another position to continue cropping.

Changing display settings for autotracts and custom tracts

Display settings that you can change include tract opacity, tract name, tract color, and fiber slab thickness (line depth).

To adjust an autotract's display settings, select and expand the desired tract card. To adjust a custom tract's display settings, click the **Display** tab on the tracts card.

- To change the 2D and 3D tract opacity, drag the slider to the desired percentage.
- To change the tract name, enter text in the text entry field next to **Name**.
- To change the tract color, select one of the predefined colors. Double click any color that is not the DEC color to open the color picker and select any color. The rainbow-colored circle is for directionally encoded colorization of fibers.
- The **Fiber Slab Thickness** controls the display of fibers that are within a slab around the current slice. The slab is the thickness of the line depth value. The fibers shown in the 2D image views will be within half of the line depth value on either side of the current 2D plane. To change line depth of a tract in the 2D views, click **[+]** or **[-]** or enter a value between 0 mm and 10 mm. The default value is 3 mm.

Deleting a tract

1. In the **Tractography** drawer, click the autotract or custom tract that you want to delete.
2. Click **[Delete]**.
3. Click **[Delete]** to delete the tract.
4. Click **[Delete]** on the dialog to confirm you want to remove the tract.

Note: Deleting the last tract automatically creates a new custom tract that is ready for you to define the start of a region of interest.

Deleting multiple tracts

To delete more than one tract, tap the More button at the top-right of the **Tractography** task.

- To delete all tracts, tap **[Delete All]**. On the confirmation dialog, tap **[Delete All]**.
- To delete multiple tracts, tap **[Delete Multiple]**. Select the Delete button next to each desired tract card. Click **[Done]** when finished.

Deleted tracts are cleared from the system.

How the Enhanced CSD tract fidelity works

The Enhanced CSD tract fidelity uses an enhanced version of the Constrained Spherical Deconvolution (CSD) algorithm. This algorithm constructs tracts using a probabilistic algorithm that takes into account multiple directions of water diffusion in each voxel. It constructs a fiber orientation distribution (FOD), which captures the probability of any given fiber direction existing in a voxel. The algorithm starts from random points (seeds) within the entire **Start ROI** and traces fibers along the directions with the highest probabilities in the FOD. The algorithm continues tracing a fiber until one of the following conditions is met:

- The probability of any given fiber direction (that is, FOD amplitude) of the next point traced in the fiber goes below the **Fiber tracking cutoff**.
- The angle between successive steps of the fiber exceeds the probabilistic **CSD Max Angle**.
- The fiber runs outside of the brain mask.
- The **Max # of Fibers** is met.

- The maximum number of seeds is exceeded.

The fiber is then automatically excluded if it falls outside the **Min Fiber Length** or the **Max Fiber Length**.

The returned tract is the collection of fibers generated from the seeds.

The Enhanced CSD algorithm keeps trying to find fibers up to the **Max # of Fibers** parameter value. Defining a mid or end ROI guides the algorithm to find more fibers going between the ROIs. When only a start ROI is defined, the algorithm finds more extraneous fibers.

Enhanced CSD parameters

Parameter name	Parameter description	Default value	Range
CSD Max Angle	Maximum angle, in degrees, between successive probabilistic steps (points) taken by the algorithm (You can increase this parameter to see more angular change and branching in the course of the tract, but this may also create more extraneous fibers that may need editing.)	7 degrees	1–15
Min Fiber Length	The minimum length of the fibers generated	30 mm	0–200
Max Fiber Length	The maximum length of the fibers generated	200 mm	10–500
Max # of Fibers	The desired number of fibers to create. The algorithm may not find the maximum number of fibers if the ROIs or other parameters are too restrictive. Also note that the number of seed points in the start ROIs is proportional to the maximum number of fibers parameter. Even if the first compute of a tract does not result in reaching the maximum number of fibers, increasing the maximum number of fibers parameter may result in more fibers being found because there are now more candidate fibers from the start ROIs.	1900	100–10000
Fiber tracking cutoff	The probability threshold for any given fiber direction (that is, FOD amplitude) that needs to be exceeded for fiber to continue to propagate (Lowering the Fiber tracking cutoff makes the algorithm more sensitive, which could result in more extraneous fibers. Raising the Fiber tracking cutoff makes the algorithm more specific by only displaying the most probable fibers.)	0.15	0.05–0.4
Seed multiplier	Increasing this parameter results in more start-seeds in the start ROI, which may result in more fibers to be found to make a tract look more dense.	16	1-5000

Guidance for adjusting tract parameters

Assess the adequacy of the tract for your intended purpose before proceeding with navigation and adjust tract parameters if needed. Varying some of the parameters will cause you to see less of a tract (for example, using a small **CSD Max Angle** may result in a smaller or less dense tract). Refer to the following table for additional information about adjusting tract parameters.

Situation	Parameter adjustments
The tract is clearly incorrect.	• Make sure all parameters are within acceptable ranges. • Make sure the correct scan protocol was used. Refer to the <i>Stealth AXIS™ Tractography Imaging Protocol</i> for more information. • Make sure that the gradient assignment is done correctly by assessing the quality of the DEC Map colorization.
The fiber density of the tract is too low.	For the Enhanced CSD parameters, try increasing the Max # of Fibers or lowering the Fiber tracking cutoff. This may create some extraneous fibers that can then be excluded with Exclude ROIs or with Crop. Note: The tract may take longer to process with the changed parameters.

Situation	Parameter adjustments
The tract is missing areas that you expected to be included.	- For the Enhanced CSD parameters, try decreasing the Fiber tracking cutoff. - Try increasing the CSD Max Angle. - Try increasing the volume of the ROIs (except Exclude ROIs) or changing their location. - Try adding another Start ROI in the deficient area. - Try increasing the Max Fiber Length. - Review the brain mask for holes where fibers will not be generated. (See “Processing diffusion images” for additional information.) Note: The tract may take longer to process with the changed parameters, and the tract may have false positives that can be edited with Exclude ROIs.
The tract has too many false, extraneous fibers.	- Try adding End, Mid, or Exclude ROIs or use Crop. - For the Enhanced CSD parameters, try raising the Fiber tracking cutoff. - Try decreasing the CSD Max Angle.
The tract has false, extraneous long fibers.	Try decreasing the Max Fiber Length or adding an Exclude ROI.
The tract has false, extraneous looping fibers.	Try decreasing the CSD Max Angle or adding an Exclude ROI.

Planning

In the **Planning** task and in the **Navigation** task, you can use a variety of tools to define surgical plans to target structures appropriate for your procedure.

After you define a surgical plan, you can change the plan in the following ways:

- Edit the target point
- Edit the entry point
- Edit the plan trajectory
- Change the plan name
- Change the plan color
- Lock the plan to prevent inadvertent changes
- Show or hide the plan
- Display a cylindrical margin around the plan

Refer to “Editing a surgical plan” for instructions on editing plans.

Warning: Verify that a plan’s target point, entry point, and trajectory are clinically valid before using it for a procedure. Traverse the surgical plan to check the entry, target, and trajectory after any edit. If you change the merge reference image after the plan is created, review your surgical plan to avoid inaccurate navigation. Some existing plan data is remerged to the new reference image and may have shifted.

In the **Planning** and **Navigation** tasks, the 3D model automatically rotates to look down the selected plan. If you prefer to set the 3D model to a fixed orientation, change the **Auto-rotate 3D View to Plan** setting to OFF on the **Navigation** tab in your surgeon profile. See “Editing a procedure.”

Defining a surgical plan

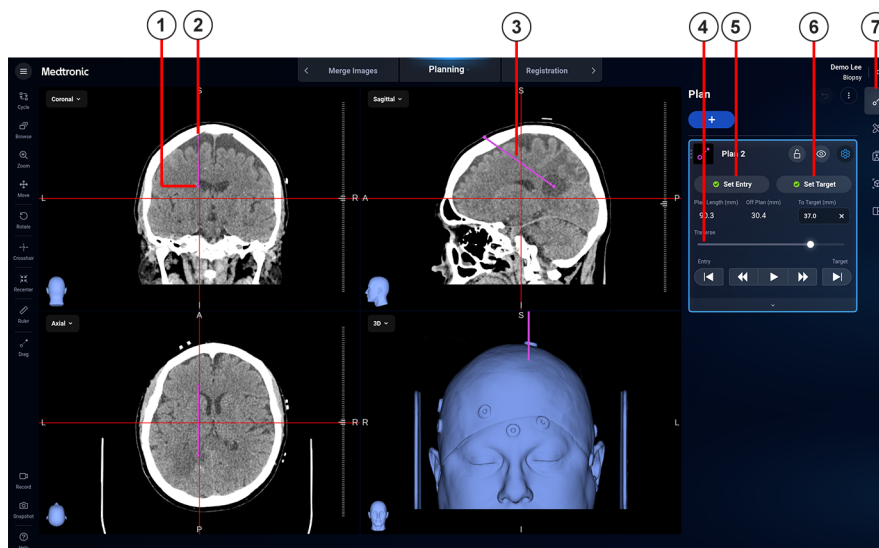
To define a surgical plan:

- Tap the Plans button to open the **Plan** drawer.
- Scroll through the 2D image slices and find the slice that contains the surgical target.
- Tap a target point in one of the 2D views. Refine its location by tapping in any of the other views.
- Tap **[Set Target]**.
- Tap an entry point on one of the 2D views or on the surface of the 3D model.
- Tap **[Set Entry]**.
- Simulate movement along the plan by doing one of the following:
 - Drag the **Traverse** slider.

- Tap the Backward and Forward buttons to move along the plan by -1.0 mm or +1.0 mm, or tap the Skip Backward or Skip Forward buttons to move the plan to the entry or target location.
- Use the left, right, up, or down arrow keys on the keyboard.
- Tap the Play button to automatically traverse the plan.

If you move a point by mistake, tap the Undo button to undo one change.

8. If you want to create another plan, tap **[+]** in the **Plan** drawer and repeat the steps above.



- 1 Plan target
- 2 Plan entry
- 3 Plan trajectory
- 4 Traverse slider
- 5 **[Set Entry]**
- 6 **[Set Target]**
- 7 Plans button

Editing a surgical plan

- Use the touchscreen to edit a target or entry by moving the crosshairs and tapping either **[Set Entry]** or **[Set Target]**.
- Tap the Drag button to drag the plan entry or trajectory and rotate the plan around the target in any 2D view.
- Tap the Drag button to drag the plan target and rotate the plan around the entry in any 2D view.
- While holding the <SHIFT> key, you can use the mouse to drag the plan entry or trajectory and rotate the plan around the target in any 2D view.
- While holding the <SHIFT> key, you can use the mouse to drag the plan target and rotate the plan around the entry in any 2D view.

Editing the target

1. Select the plan you want to edit from the **Plan** drawer.
2. Tap the Skip Forward button in the drawer to go to the target and view the current position.
3. In any view, tap the location of the desired target.
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap **[Set Target]**.
The clinical application sets the target to the point indicated by the crosshairs.

Editing the entry

1. Select the plan you want to edit from the **Plan** drawer.
2. Tap the Skip Backward button in the drawer to go to the entry and view the current position.
3. In any view, tap the location of the desired entry.
To magnify the entire image, tap **[Zoom]**, and then drag with one finger up to zoom in or down to zoom out.
4. Tap **[Set Entry]**.
The clinical application sets the entry to the point indicated by the crosshairs.

Editing the plan trajectory

Follow these steps to edit a plan's trajectory to pass through or avoid structures.

1. Select the plan you want to edit from the **Plan** drawer.
2. To review your plan trajectory, traverse the plan using one of the controls in the drawer:
 - Tap the Play button to have the clinical application move along your plan.
 - Use the **Traverse** slider to manually move along your plan.
3. While holding the <Shift> key and using the mouse, or tapping the Drag button and using your finger, you can drag the plan trajectory in any view. Select a view that allows you to visualize the relevant structures. You can only drag a plan in 2D Views, not in 3D Views.
Note: Editing a plan's trajectory will change either the planned entry or target.

Locking plans

Follow these steps to lock a plan to prevent changes.

1. In the **Plan** drawer, tap the plan you want to lock.
2. Tap the Lock button to lock the selected plan.
3. To unlock a plan, tap the Lock button again.

Changing plan names and colors

1. Select the plan you want to edit from the **Plan** drawer.
2. To change the plan name, tap the Settings button, and then tap the Expand button to enter text in the **Name** field.
3. To change the plan color, tap the Expand button, and then select one of the predefined colors, or double-tap a predefined color and select a custom color.

Displaying a margin around the plan

1. Select the plan you want to edit from the **Plan** drawer, and then tap the Settings button.
2. Tap the Expand button.
3. Turn on the **Safety Margin**.
The default diameter for the safety margin is 5 mm.
To change the diameter, tap the **[+]** or **[-]** on the diameter slider to edit the value. You can also enter a value in the diameter text field or use the slider bar.

Deleting a surgical plan

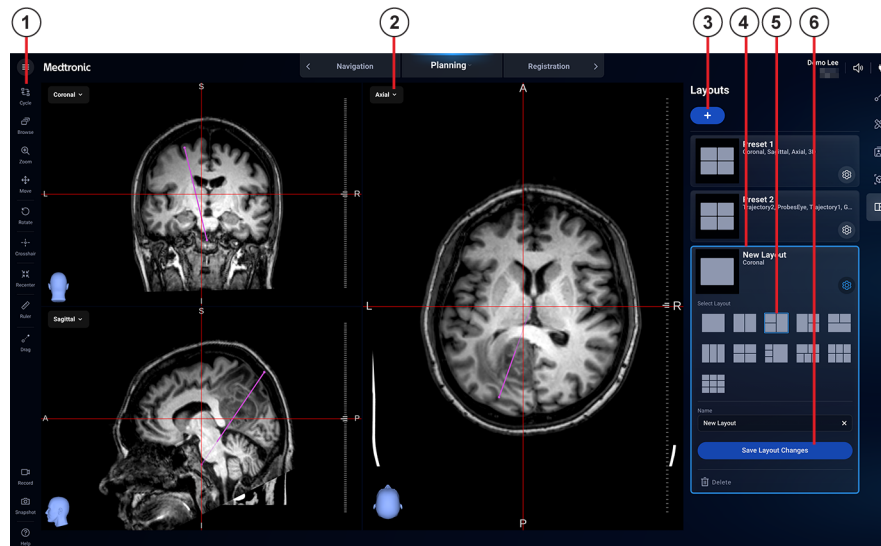
To delete a plan do one of the following:

- Tap the Settings button, tap the Expand button, and then tap **[Delete]**.
- Swipe right and then tap the Delete button.
- Tap the More button and then tap **[Delete Multiple]** or **[Delete All]**.

Adjusting layout configurations

The default image view layouts are based on the selected procedure and task (for example, four quadrants that display the **Axial**, **Sagittal**, **Coronal**, and **3D** views). You can select one of the preconfigured layouts, modify an existing layout and save it, or create your own configuration. Layouts you modify or create are saved to the current surgeon profile and selected procedure.

- To cycle through the saved layouts, tap **[Cycle Layouts]**.
- To display a different screen layout, tap one of the layout cards in the **Layouts** drawer.
- To temporarily change a multiple view layout to a one view layout, double-tap the Image Selection drop-down menu in any view. Double-tap again to return to the previous layout.
- To change the display of a view, select the desired view from the Image Selection drop-down menu.
- To save a layout view configuration that can be reused:
 1. Tap the Layouts button.
 2. Tap the Settings button on an existing layout to edit it, or tap **[+]** to create a new layout configuration.
 3. Tap a **Select Layout** card.
 4. To select a different view type, tap the Image Selection drop-down menu in the desired view.
 5. Tap **[Save Layout Changes]**.
 6. Name your layout and tap **[Confirm]**.



- 1 **[Cycle Layouts]**
- 2 **Image Selection** drop-down menu
- 3 Create a new layout configuration
- 4 Layout view
- 5 One of the layout buttons
- 6 **[Save Layout Changes]**

Navigating

To navigate, you must match the patient's exam to the patient's anatomy using a process called registration. See the Stealth AXIS™ Cranial clinical application instructions for use for information about setting up equipment, verifying instruments, performing a registration, and navigating. If you know autotracts have been generated, but the clinical application indicates none have been generated, go back to the **Tractography** task and verify each autotract you want to display. During navigation, the clinical application updates the 2D views in real time to reflect changes in the position of the surgical instrument's tip on the anatomy.

Warning: Frequently confirm navigation accuracy and system responsiveness during live navigation. Use the navigation instrument to touch several bony anatomical landmarks and confirm that the locations identified on the images match the locations touched on the patient. If accuracy degrades, re-register the patient.

Exporting patient data

Follow these instructions to export patient data (including images, merges, and plans), snapshots, or video to media or network locations:

Note: The system is not meant for long-term data storage. Patient data is to be transferred to the system for the patient case. After case completion, data can be transferred to a PACS or another customer-managed storage device for long-term storage, or the data can be deleted. Data transfer and handling is managed by the customer.

1. Tap the Menu button and tap **Export**.
2. Select one or more patients whose data you want to export.
3. Select an export destination, such as a USB, CD/DVD-ROM, the filesystem, PACS, or another configured Stealth AXIS™ system from the **Destination** drop-down menu.
Select a location for the export, if necessary.
See “Adding a PACS server” to set up a PACS server.
4. Tap the tab for the type of data you want to export.
 - Tap the **All Patient Data** tab to export all data associated with the patient (for example, CT and MR images, snapshots, and videos) and planned data (for example, annotations, plans, and merge data).

Note: This selection archives CT and MR images in the Stealth format. Select the options described in the next step to modify aspects of the exported data.

Note: To view and edit snapshots before exporting, see “Viewing snapshots.”

- Tap the **Snapshots and Videos Only** tab to export snapshots and videos only.
 - Tap the **Patient Coordinate File Only** tab to export only the patient coordinate file.
5. Select the desired export options. The options that appear depend on the data type selection and the data that exists for the selected patients.

DICOM

- If desired, tap the **Description** field and change the text. This text appears in the Patient Name column of the export.
- To anonymize the export, turn on the **Anonymize** toggle and enter prefix text in the **Anonymization Prefix** field. When used, this text replaces the Description text in the Patient Name column of the export and changes patient identifying information in the DICOM tags of the export.

Warning: Use care if you are planning to use the patient images for tractography because the anonymizer may delete DICOM tags necessary for tractography.

Note: Anonymizing does not delete identifying information in the images themselves, such as user-entered identifying information in any text input fields.

Note: Snapshots are not anonymized.

- Select **Preserve Series Description** to retain the text in the **Description** field in the export. Clear the option to change the Series Description of the export to the entered **Anonymization Prefix**.
- To add an optional accession number attribute to the export, turn on the **Define Accession Number** toggle and enter an accession number in the **Accession Number** field.
- Select **Include Original DICOM Format Images** to add the original DICOM images to the patient data export. When the **Anonymize** toggle is on, the original DICOM image files are anonymized.

File Options

- Select **Include Ultrasound Sequences** to include ultrasound sequences, in mp4 format, in the export.
- If the selected patients have snapshots or videos, they will be included in the export by default, in jpg and mp4 formats respectively. Turn off the **Include Snapshots and Videos** toggle to exclude these files. If you also want to include snapshots and videos in DICOM format, select **Add Snapshots and Videos in DICOM Format**.

Patient Management

Select **Delete Patient After Successful Export** to automatically delete patient data when the export successfully completes.

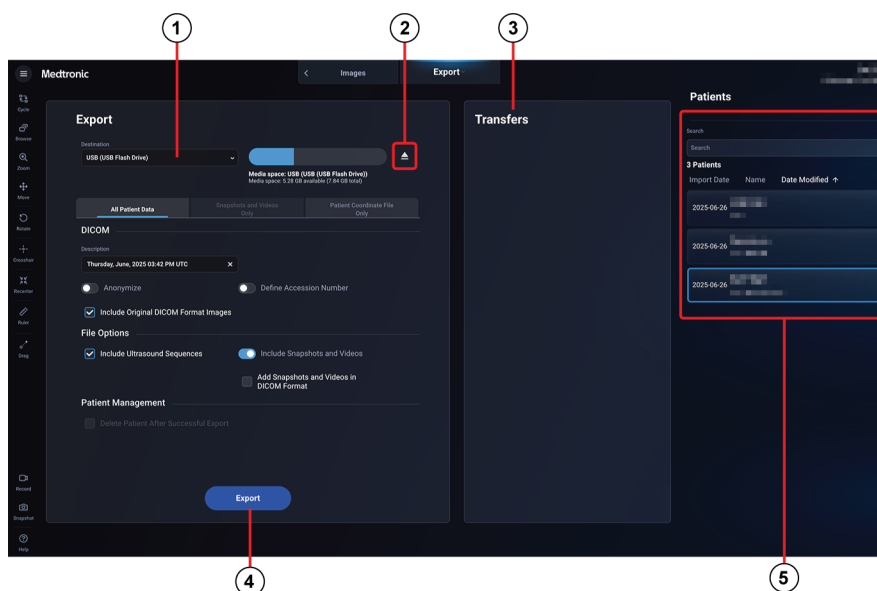
6. Tap **[Export]**.

The **Transfers** list shows export progress.

- Tap **[Cancel]** to stop an export that is in progress.
- Tap **[X]** to close a patient name whose transfer is complete.

Unfinished transfers will be canceled if you navigate away from the **Export** task while an export is in progress.

Note: When you export to a USB drive, the Eject button is displayed on the clinical application screen. After the file transfer is complete, tap the Eject button. If the USB drive has a light, wait for the light to stop flashing before you remove the USB drive from the system.



- 1 Destination drop-down menu
- 2 Eject button
- 3 Transfers
- 4 [Export]
- 5 Patients

Viewing snapshots

Snapshot viewer allows you to view and edit snapshots before exporting them.

1. Tap the Menu button.
2. Tap **[Patient Admin]**.
3. Tap the **Snapshots** tab.
4. Double-tap the snapshot that you want to view or edit. The snapshot enlarges.
 - To rename the snapshot, enter a new name in the **Rename** field, tap **[Rename]**, and then tap **[Confirm]**.
 - To delete the snapshot, tap **[Delete]**.

Managing patients and images

To clone, move, delete, or combine selected patient information, use the **Patient Admin** task. Follow these steps to open the **Patient Admin** task:

1. Tap the Menu button.
2. Tap **[Patient Admin]**.

The **Patient Admin** task displays.

Cloning patients

Cloning a patient includes images, planned data, and any modifications you have made during a session.

Follow these steps to clone selected patients in the **Images** task:

1. Find the desired patient in the **Patients** list.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. If the patient card is not selectable, it is currently in use by someone else. Tap the Clone button.
Note: This is only applicable when using the planning server.
3. If the patient card can be selected, it is not currently in use. Select the patient card to expand it and tap **[Clone Patient]**.
4. On the **Clone Patient** dialog, enter a unique patient name and select the check boxes to include edit history, snapshots, and videos.
5. Tap **[Clone]**.

The new patient card displays in the **Patients** list.

Note: You cannot recombine edits on a cloned patient with the original patient.

Combining patients

Follow these steps to combine two or more selected patients in the **Patient Admin** task:

1. Select patients to combine.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. Tap **[Combine Selected Patients]**.
3. Select the target patient into which the selected patients will be combined.
4. Tap **[Combine]**.

Renaming patients

Follow these steps to rename selected patients in the **Patient Admin** task.

Note: If a patient is used in the current procedure, that patient cannot be renamed.

1. Select a patient.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. Tap the Settings button next to the patient name.
3. In the **Name** field, enter the new patient name.

Deleting a patient

Follow these steps to delete a patient from the system.

Note: If a patient is used in the current procedure, that patient cannot be deleted.

Note: All data associated with the patient, including videos and snapshots, are deleted.

In the **Patient Admin** task:

1. Select a patient.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. Tap the Settings button to expand the card.
3. Tap **[Delete]**.
4. Tap **[Delete]** on the dialog to confirm you want to remove the patient data from the system.

The patient data is removed from the system.

Deleting multiple patients

In the **Patient Admin** task, follow these steps to delete more than one patient:

Note: If a patient is used in the current procedure, that patient cannot be deleted.

Note: All data associated with the patient, including videos and snapshots, are deleted.

1. Tap the **More** button at the top right of the **Patients** list.
2. To delete all patients, tap **[Delete All]**. On the confirmation dialog, tap **[Delete All]**
3. To delete multiple patients, tap the **[Delete Multiple]** button. In the Patients dialog, select the Delete button next to each desired patient. Click **[Done]** when finished.

All patients and patient data are cleared from the system.

Moving images from one patient to another

Follow these steps to move images for selected patients in the **Patient Admin** task:

1. Select patients.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. In the **Images** tab, tap the patient images that you want to move.
3. Tap **[Move Selected Images]**.
 - a. To move the images to a new patient, enter a new patient name and MRN.
 - b. To move the images to an existing patient, select the patient name from the **Patients** drop-down menu.
4. Tap **[Move Selected Images]**.

Deleting images

Follow these steps to delete images for selected patients in the **Patient Admin** task.

Note: Deleting the only image for a patient deletes the patient.

1. Select patients.
Note: To search for a patient, enter the patient name or MRN into the text field.
2. In the **Images** tab, tap the patient images that you want to delete.
3. Tap **[Delete Selected Images]**.
4. Tap **[Delete]** to confirm.
Note: All data associated with the patient, including videos and snapshots, are deleted.

Exiting a case

Follow these instructions to exit a case:

1. Tap the Menu button.
2. Tap the Home button.

The **Select Procedure** screen displays.

Exiting the clinical application

Follow these instructions to exit the clinical application:

1. Tap the Menu button.
2. Tap the Power button.
3. Tap **[Log Out]**.
4. Tap **[Log Out Now]**. Tapping **[Cancel]** will abort the action.

The login screen displays.

Note: If you do not tap **[Log Out Now]**, the system will automatically log out after 25 seconds.

Shutting down the system

Follow these instructions to shut down the system:

Caution: Ensure that the workstation cart is shut down using the clinical application. Shutting down the system by disconnecting from an electrical outlet and letting the batteries drain or holding the power button for more than 6 seconds is not recommended and could damage the workstation cart or put the cart into a condition that requires maintenance.

Once logged into the clinical application, to shut down the system:

1. Tap the Menu button.
2. Tap the Power button.
3. Tap **[Power Down]**.
The clinical application prompts you to confirm.
4. Tap **[Power Down Now]**. Tapping **[Cancel]** will abort the action.
The system will fully shut down.

Note: If you do not tap **[Power Down]**, the system will automatically shut down after 25 seconds.

To power down the system from the login screen:

1. Tap the Power button in the lower-right corner to access the drop-down menu.
Note: The Power button does not display during remote planning sessions.
2. Tap **[Power Down]**.
3. The system prompts you to confirm.
4. Tap **[Power Down Now]**. Tapping **[Cancel]** will abort the action.
5. The system will fully shut down.
Note: If you do not tap **[Power Down]**, the system will automatically shut down after 25 seconds.

Restarting the system

Follow these instructions to restart the system.

1. Navigate to the login screen by following the instructions in “Exiting the clinical application.”
2. Tap the Power button in the lower-right corner to access the drop-down menu.
Note: The Power button does not display during remote planning sessions.
3. Tap **[Restart]**.
The system prompts you to confirm.
4. Tap **[Restart Now]**. Tapping **[Cancel]** will abort the action.
Note: If you do not tap **[Restart Now]**, the system will automatically restart after 25 seconds.

Troubleshooting

Clinical application icons

If you see one of the following icons on the clinical application screen, review the information displayed next to the icon on the screen.



Tap the icon to display additional information. Tap **[Close]** to hide the information.

Image import problems

Find the symptom in the table below and follow the troubleshooting steps:

Symptom	Troubleshooting steps
You imported a series and slices are missing in the area of anatomical interest.	• If you imported the series from a PACS, ask Radiology to make sure that all slices are present in the original scan. • Ensure that slice spacing and thickness remain the same throughout the scan. Sometimes gaps occur because the spacing and thickness changes in the scan based on how radiology acquires or processes the scan. • If you imported the series from media, use a DICOM viewer to make sure that all slices are present in the original scan. • If all slices are in the original scan, delete the imported series and reimport it. To delete the series, see “Deleting images.” • If the problem persists after you reimport the series, contact Medtronic technical services.
O-arm™ system images contain distortion.	Make the following changes and then rescan the patient: • Remove metal and nonessential objects from the scanner’s field of view (for example, remove instruments, extra tubing, and extra drapes). • Adjust the scanner or the patient to center the region of interest in the scanner’s field of view and include as much of the patient’s anatomy as possible within the scan.
The images do not contain the entire anatomical region of interest needed for the procedure.	Adjust the scanner or the patient so that as much as possible of the region of interest is in the scanner’s field of view and then rescan the patient.
You need to correct the orientation of a series due to mislabeling in the scanner.	Tap [Correct], then tap the options to correct the orientation. Note: When you select a diffusion image series, [Correct] is disabled because the image orientation cannot be changed.
The level and width adjustments seem too sensitive (for example, they change from very light to very dark, and it may be difficult to show defined anatomy).	1. In the Images task, tap the arrow under Series to display details about the patient series, including the original bit depth. 2. In the Patient Images list, tap the Settings button on the patient series. 3. Tap the 2D tab. 4. Tap the Bit Depth drop-down menu and select a bit depth that is lower than the original. 5. Adjust the Level and Width sliders.
You need to move a series from one patient name to another.	See “Moving images from one patient to another” for details.
You need to change the modality of a series.	Select the patient series in the Patient Images list, tap the Settings button, and select the correct modality.
A warning message that includes: “Minimum of three slices required.”	Ask radiology to review the scan to comply with the Imaging Protocol.
A warning message that includes: “This exam is restricted from use in the tasks below: Registration.”	If you want to include the series in image views for planning or navigation, you can manually merge the series with a series that is acceptable for registration.
A warning message that includes: “This may affect functionality or navigation accuracy of the system.”	Ask radiology to reformat the scan to comply with the <i>Stealth AXiS™ Imaging Protocols for Navigated Procedures</i> .
A warning message indicates an issue about slice thickness or slice spacing.	Ask radiology to reformat the scan to comply with the <i>Stealth AXiS™ Imaging Protocols for Navigated Procedures</i> .
A message indicates the exam size is too large.	Ask radiology to reformat the scan to comply with the <i>Stealth AXiS™ Imaging Protocols for Navigated Procedures</i> .
A message indicates an issue with the field of view (FOV).	Ask radiology to reformat the scan to comply with the <i>Stealth AXiS™ Imaging Protocols for Navigated Procedures</i> .
A message indicates an issue with gantry tilt.	Ask radiology to reformat the scan to comply with the <i>Stealth AXiS™ Imaging Protocols for Navigated Procedures</i> .
A warning message that includes: “Diffusion import failed.”	Ask radiology to reformat the scan to comply with the <i>Stealth AXiS™ Tractography Imaging Protocol</i> .

Symptom	Troubleshooting steps
A warning message that includes: "Diffusion gradient directions are not well distributed."	Uniformly distributed diffusion-encoding directions are recommended for diffusion scans. Ask radiology to reformat the scan to comply with the <i>Stealth AXIS™ Tractography Imaging Protocol</i> . If using a Philips scanner, avoid Overplus On.
A warning message indicates the B value is not in the optimal range.	The optimal B value range for the Enhanced CSD fidelity is 1000 s/mm² – 3000 s/mm². If the B value is within the optimal range for one tract fidelity, select that tract fidelity before tapping [Compute Tract] in the Tractography task. If the B value is not within the optimal range for either tract fidelity, the clinical application may process the diffusion scan, but the resulting fiber tracts may not be optimal.
A warning message indicates the series contains less than 12 gradient directions.	For the Enhanced CSD tract fidelity, the clinical application requires at least 12 unique gradient directions to create tracts. If the clinical application finds more than one unique non-zero B value (also known as a shell) within the diffusion gradient series, the clinical application only selects a single shell for use. The number of unique gradient directions in the selected shell must be greater than or equal to 12.
A message indicates that the diffusion modality is not available for Registration.	If you want to include the diffusion series in image views for planning or navigation, you can merge the series with a series that is acceptable for registration.
A message indicates that DICOM tags are missing.	Anonymizers and pre-processing features in the clinical application may remove DICOM tags that are necessary for tractography. Make sure that any anonymizer or pre-processing clinical application you use is not removing or modifying the tags that are listed as missing.
When you try to search for images on a PACS, a warning message displays.	After each of the following steps, try your search again to determine whether the step resolved the issue: 1. Make sure the network cable is plugged in on both ends if using a wired configuration. 2. If using wireless connectivity, ensure that an adequate signal is available. 3. See "Running DICOM tests." 4. Check the network settings in the Admin application: a. Tap the Menu button. b. Tap [Admin]. 5. Make sure the ethernet port at the facility has an active network connection. 6. Make sure there are no other devices on the facility's network that are using the same IP address and network configuration. If the problem persists, contact Medtronic technical services.

Merge tips

When you merge images, the merged set is always used in the **Tractography**, **Planning**, **Registration**, and **Navigation** tasks.

Goal	Steps
To view only one of the merged images in the Tractography , Planning , or Navigation tasks	Tap [Images] , then tap the Show button on all the images that you do not want to view. When the Show button has a line through it, that image is not visible.
To remove an image from the merged set	Go to the Images task and turn off the Select toggle on the images that you want to remove. Note: If you deselect the image that was previously the reference image, your planning data associated with that image will be removed.

Troubleshooting in the Planning task

Find the goal in the table below and follow the steps:

Goal	Steps
To move a target (end point), trajectory, or entry point in the 2D image views	Use the touchscreen to edit a target or entry by moving the crosshairs and tapping either [Set Entry] or [Set Target] .
To display dots instead of crosshairs in 2D views	The crosshair can appear as a dot or a cross in 2D views. Use one of the following methods to change the appearance. • Select the Crosshairs button in the Image Tools sidebar. Select the display type that you want to use. • Tap the Menu button, and then tap [Preferences]. Select the procedure. Tap the Views tab. Under Crosshair Visibility, select Cross or Dot. • Press <F4> on the keyboard to cycle through the options.

Battery backup

The workstation cart will attempt to run on backup battery power when it is not plugged into an electrical outlet. An audible indicator will sound when the system is running on battery backup, and it will increase in frequency when the battery is low. The battery status icon is displayed in the upper-right corner of the clinical application screen. After a procedure is selected, tap the battery status icon to open the **System Status Menu** and view more information about the battery status. The **System Status Menu** can be accessed during any task except while in the **Procedure Select** task.

One of the following battery icons displays in the upper-right corner of the clinical application screen:

Battery icon	Color	Description
	White	The workstation cart is plugged into an electrical outlet and is charging.
	White	The system's battery backup is in use and has a charge greater than 50%.
	White	The system's backup battery is in use and has a charge greater than 25% and less than 50%.
	White	The system's backup battery is in use and has a charge greater than 10% and less than 25%.
	Flashing orange	The system's battery backup is in use and has a charge less than or equal to 10%.
	Flashing red	The system's battery backup service has detected an error or the workstation cart battery has no charge. Unplugging the system's power cord from the electrical outlet may result in immediate system shutdown. Contact Medtronic technical services.

Note: Battery icons do not display in the clinical application on the Planning Station or in a remote planning session.

Accessing remote support

Remote Support is a digital service that enables Medtronic to provide features such as: remote diagnostics, subcomponent level diagnostic alerts, and remote troubleshooting support using a real-time-screen-share feature. Remote Support requires an active, secured Internet connection between the Covered Product and Medtronic for connectivity. The screen-share function requires the person operating the system to initiate a screen-share session by physically opening the Remote Support application with the assistance of Medtronic Technical Support.

If requested to do so by Medtronic technical services, access remote support. For detailed information about accessing a remote support session, see the *StealthAXiS™ IT Guide*.

Note: Remote support is unavailable when using the remote planning feature.

Appendixes

A: Transition guide for StealthViz™ clinical application users

The following table includes information about differences between some tasks in the StealthViz™ clinical application and the tractography feature in Stealth AXIS™ Cranial clinical application.

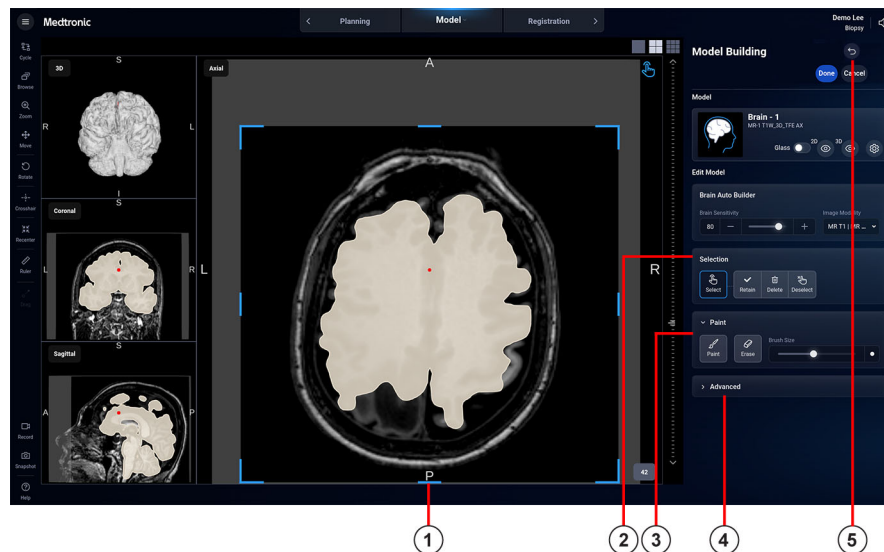
Task	StealthViz™ clinical application	Stealth Tractography in the Stealth AXIS™ Cranial clinical application
Diffusion exam processing	• Initiates diffusion processing after loading exams (in the View tab). • Computes tensor volumes (DTI). • Includes optional motion correction.	• Initiates diffusion processing in the Images task as you are deciding which exams to load. • Computes tensor volume iFOD (Enhanced CSD). • Includes optional motion correction and eddy current correction.
Defining tracts: Regions of Interest	• Allows one each of start, mid, and end ROIs. • Supports box shapes and 3D model-based ROIs. • For box-shaped ROIs, clicking a view drops a default box. Click and drag to resize or rotate the box or hold the <Shift> key and click and drag to move the box (in either 2D or 3D views).	• Allows multiple of each of start, mid, end, and exclude ROIs. • Supports box shapes and freeform painted shapes (no 3D model-based ROIs). • For box-shaped ROIs, click and drag in a 2D view to define the placement and size of box in one action, and then click and drag to move, resize, or rotate (in 2D views only). ROI shapes are visible but not editable in 3D views. • For freeform ROIs, multiple disjointed areas can be considered as one ROI.
Tract parameters: Seed Density	Uses a default of 1 seed/voxel and allows a maximum of 5 seeds/voxel.	For Enhanced CSD, the number of lines desired is specified in Parameters , so adding mid and end regions helps make the tract denser, as more fibers are found along the designated path.
Tract editing: Crop and Exclude	• Allows crop with a lasso draw in either 2D or 3D views around fibers – in either whole-fiber or scissor mode (crop action is undone by a tract recompute). • Does not include a persistent exclude feature.	• Allows crop with a Box draw in the 3D view around fibers – whole length of a partially overlapped fiber is removed (crop action is undone by a tract recompute). • Allows multiple Exclude ROIs in the 2D views. The effect of these exclude regions persists across tract recomputes.
Use of fiber tracts in navigation	• Requires export to the Stealth AXIS™ Cranial clinical application. • Allows either geometry (tube like) or “labels” (3D objects) style. • Only supports constant color of fiber tracts once in the Stealth AXIS™ Cranial clinical application (no DEC coloring).	• Does not require export or transfer. Uses tracts to navigate in the same clinical application. • Retains fiber-like look, with either constant color or DEC coloring. Fiber tract opacity can be varied in both 2D and 3D views. • Allows export to another Stealth AXIS™ system.
“Hybrid” output	Exports tracts embedded into a DICOM dataset as bright white areas.	Does not include a hybrid output feature.

B: Building and editing models

Model editing tools

When building and editing 3D models, you can use a variety of editing tools for each model.

Editing option	Behavior
Selection card	Displays the buttons for retain, deselect, and delete options. - To retain only a portion of the 3D model, tap the area you want to retain and then tap [Retain]. - To delete a portion of the 3D model, tap the area you want to delete and then tap [Delete]. - To cancel the selection of the 3D model, tap [Deselect].
Paint card	Displays the buttons for painting and erasing within a 2D image. - To use the [Paint] and [Erase] tools, tap [Paint]. - Swipe a 2D image view into the main window, adjust the Brush Size slider, and then drag on the image to paint or erase.
Advanced card	Refines the model with additional options including: - To fill small gaps or to smooth the model, tap the [+] and [-] to adjust the Gap Fill and Smooth sliders. - To reduce the model's size, tap [Contract]. - To increase the model's size, tap [Expand]. - To delete items that are not connected to the model, tap [Remove Islands]. - To fill holes in the model, tap [Fill Holes].
Cropping square	Removes the outside edge of a model. - Swipe a 2D image view into the main window and then adjust the cropping square to surround only the portion you want to keep. The model updates automatically when you adjust the cropping square. - To adjust the cropping square, drag the cropping line to the preferred position. You can adjust the cropping square in any 2D view.
Undo button	Undoes all generic tool actions, all threshold sliders, and Tumor/Other Seed and 3D Paint tools.



- 1 Cropping square
- 2 **Selection** card
- 3 **Paint** card
- 4 **Advanced** card
- 5 Undo button

Building and editing a tumor or other model

Follow these steps to build a tumor model, ventricle model, or any other model type.

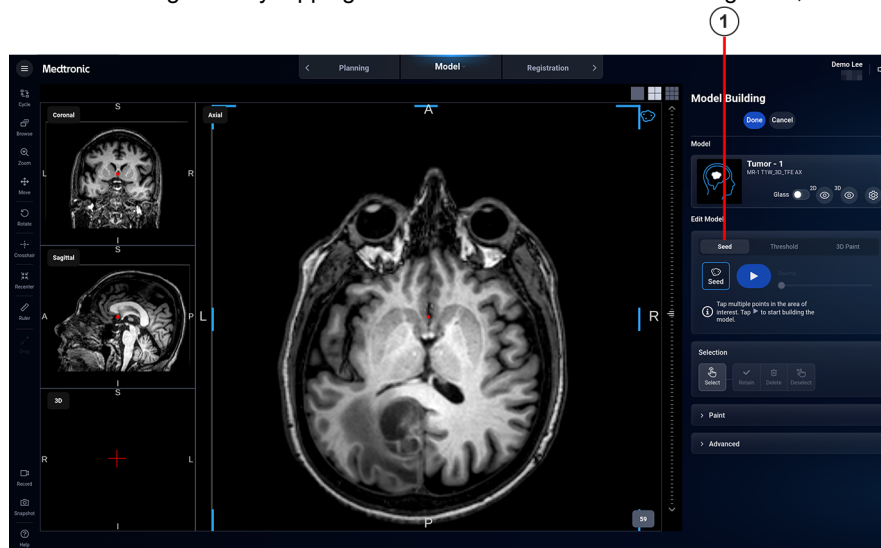
1. In the **Planning** task or **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Tumor/Other]**.
4. Select an image.
5. The **Model Building** task opens. Select one of the three methods to build a model: **Seed** (default), **Threshold**, or **3D Paint**.

Note: The clinical application builds the model using only one method. If you select a different method, the current model will be deleted.

The **Seed** tab is the preferred method to build a model. Select seed points and grow the model from those points.

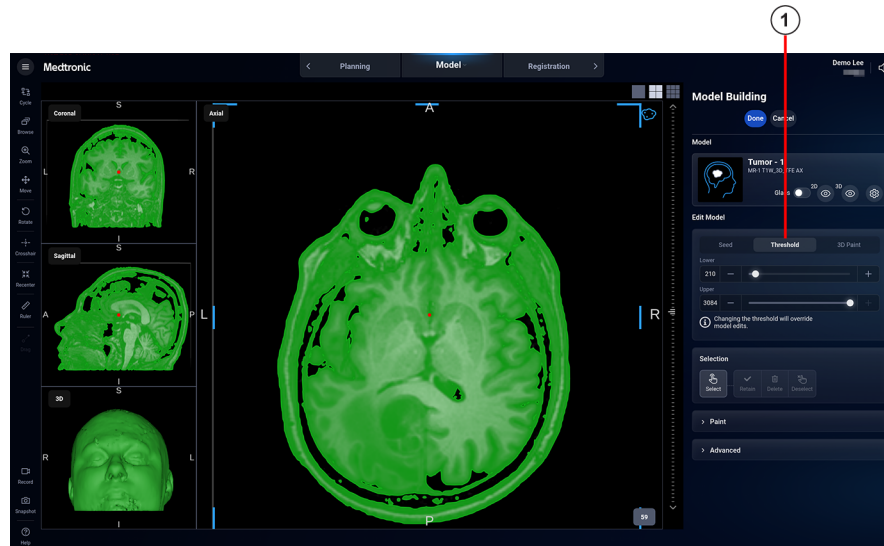
- a. Tap the **[Seed]** button.
- b. Locate a 2D image slice that contains the area of interest.
- c. Tap at least one point in the area of interest.
- d. Locate another slice that contains the area of interest and tap at least one point.
- e. Repeat the steps until you have selected at least three points on different slices.
- f. Tap the Play button to create the model.

Note: You can freeze seed growth by tapping the Pause button. To rewind seed growth, use the **Rewind** slider.



1 Seed tab

Tap the **Threshold** tab to build a model based on light and dark intensity values in the images. Use the **Lower** and **Upper** slider to adjust the thresholding.



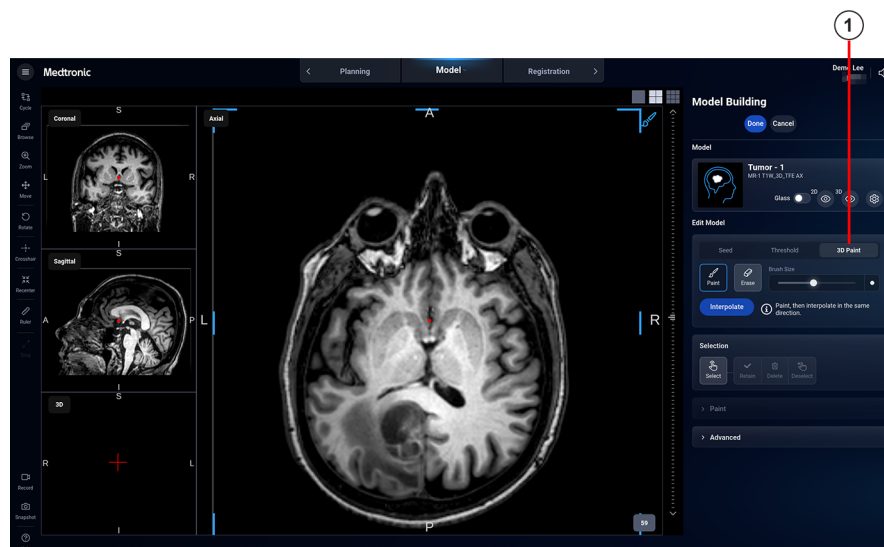
1 Threshold tab

Tap the **3D Paint** tab to build a model by painting on multiple image slices that contain the area of interest:

- Tap the **[Paint]** tool.
- Locate the image slice with the smallest portion of the area of interest visible.
- Drag the cursor on the image to paint the area of interest.
- Repeat this process to paint the area of interest on multiple image slices. You must paint in image slices of the same perspective, using only all sagittal, coronal, or axial slices.

Note: Drag on the scroll bar to scroll through the image slices. You can skip some image slices in which the anatomy is similar.

- Tap **[Interpolate]** to create the model.



1 3D Paint tab

- To display the model in 2D or 3D views in the **Planning** and **Navigation** tasks, tap the 2D or 3D Show button.

Building and editing a brain model

Follow these steps to build and edit a 3D brain model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Brain]**.
4. Select an image.
The **Model Building** task opens, and the clinical application automatically builds a brain model.
5. Edit using brain specific edit options, as appropriate:
 - To change the image modality, select an option from the **Image Modality** drop-down menu.
 - To show more or less brain anatomy in the brain model, drag the **Brain Sensitivity** slider.
6. Continue editing the model with the generic editing tools as necessary.
7. Tap **[Done]**.

Building and editing a bone model

Follow these steps to build and edit a 3D bone model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Bone]**.
4. Select an image.
The **Model Building** task opens, and the clinical application automatically builds the bone model using the **Threshold** method.
5. Use the **Lower** and **Upper** sliders to adjust the threshold.
6. Edit the bone model as necessary.
7. Tap **[Done]**.

Building and editing a skin model

Follow these steps to build and edit a skin model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Skin]**.
4. Select an image.
The **Model Building** task opens, and the clinical application automatically builds a skin model using the **Skin** method.
5. Tap **[Refine]**, and drag the **Threshold** slider to adjust the threshold.
6. Continue editing the model as necessary.
7. Tap **[Done]**.

Building and editing a vessel model

Follow these steps to build and edit a 3D vessel model.

1. In the **Planning** task or the **Navigation** task, tap the Models button.
2. Tap the Add button.
3. Under **Select Model Type**, tap **[Vessel]**.
4. Select an image.
The **Model Building** task opens.
5. Edit using vessel specific edit options, as appropriate:
 - To change the image modality in the **Vessel Auto Builder** method, select an option from the **Image Modality** drop-down menu.
 - Drag the **Threshold** slider to adjust the threshold.
 - To show more or less brain anatomy in the brain model, drag the **Brain Sensitivity** slider.

6. Continue editing the model with the generic editing tools as necessary.
7. Tap **[Done]**.

C: Adjusting views for images and models

Adjusting level and width

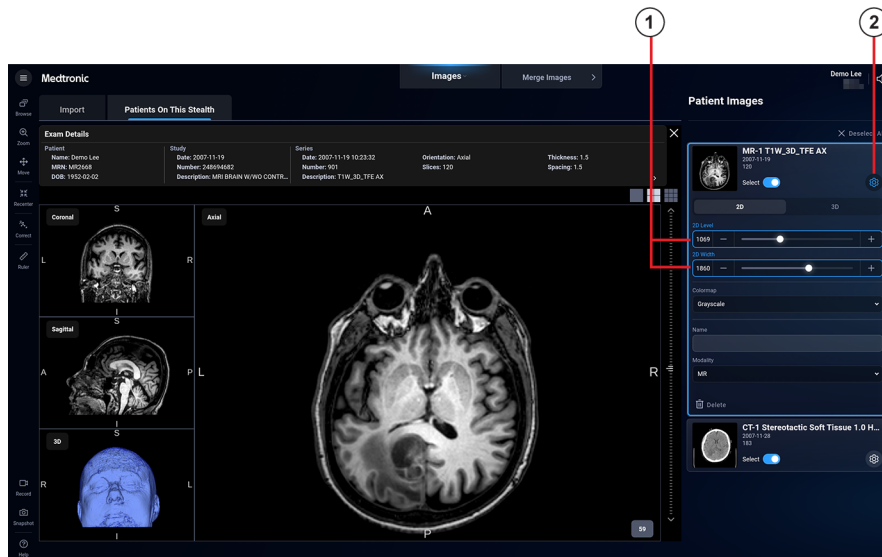
You can use the **Level** and **Width** controls to change how patient images and models display in 2D and 3D views.

Follow these steps to adjust level and width:

1. Tap the Images button to open the **Images** drawer or tap the Models button to open the **Models** drawer.
2. Tap the Settings button on the image or model you want to adjust.
3. Tap the **2D** or **3D** tab.
4. Drag the **Level** and **Width** sliders.

While the **Level** and **Width** controls are selected, you can also adjust level and width by dragging on the views.

 - Drag up and down to adjust **Level**.
 - Drag left and right to adjust **Width**.
5. Tap the Settings button or select a different image or model to deselect the level and width controls.



- 1 Level and Width sliders
- 2 Settings button

Displaying multiple images in 2D views

If you merged images, you can display multiple images in 2D views in the **Planning** and **Navigation** tasks. Tap the Images button to open the **Images** drawer, and then follow these instructions to adjust how images display:

- To manually display images on top of one another, follow these instructions:
 - Tap the 2D Show button to display or hide an image.
 - Use the **Opacity** slider to adjust the opacity of an image so that the bottom images can be seen.
- To reorder images, tap on the left edge of the image card and drag up or down in the **Images** drawer.

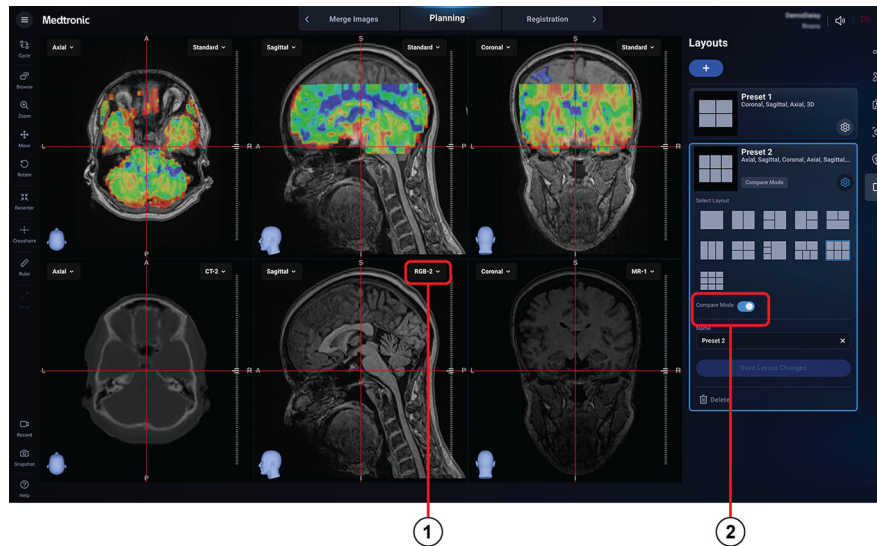
Note: Transparent images placed on top of opaque images will show both images. Images underneath opaque images will not be visible.

Using the compare mode

To compare images, follow these instructions:

1. Tap the **Layouts** button to open the **Layouts** drawer.
2. Tap the **Compare Mode** toggle button on. The view layout changes to a 6-up view with Axial, Sagittal and Coronal views on both the top and bottom rows. If desired, the view layout and view types can be changed.
3. In each view, tap the **Select Image** drop-down menu to change the desired image to display.
4. To save the layout and images selected in each view:
 - Tap in the **Name** field and type a unique preset name.
 - Tap **[Save Layout Changes]** on the compare mode preset card.

The layout is saved for the surgeon profile and procedure in use.



- 1 Select image drop-down menu
- 2 Compare mode toggle

Adjusting colormap or modality of images in 2D views

1. Tap the **Images** button to open the **Images** drawer.
2. Tap the **Settings** button on the image that you want to modify.
3. To adjust the colormap, open the **Colormap** drop-down menu and then select an option.
4. To adjust the image modality, tap the **Settings** button and select a modality from the drop-down list.

The image modality option is only available in the **Images** task.

Adjusting how models display in 2D views

For a 2D view of a model, you can choose to view contours (outlines) or to fill the outlines.

1. Tap the **Models** button to open the **Models** drawer.
2. By default, models are displayed as outlines. Tap the **[2D Fill]** toggle to fill the outlines.

The **2D Opacity** slider now affects the opacity of the fill. The outline remains as is.
3. To return to the default view, turn off the **[2D Fill]** toggle.

The **2D Opacity** slider now affects the opacity of the outline.

To reorder models, drag the models up or down in the **Models** drawer. Transparent models placed on top of opaque models will show both models. Models underneath opaque models will not be visible by tapping and dragging left edge of the card.

Adjusting how models and images display in the 3D view

If you want to change how models and images display in the 3D view, follow these instructions.

1. Open the **Models** drawer in the **Planning** or **Navigation** task.
2. Tap the Settings button on a model or image to open its display options.
3. Edit the attribute fields to change how the model displays in the 3D view.

Editable fields for models:

- **Name**
- **ColorMap**
- Select **Shaded**, **Non-Shaded**, or **Glass**.

Editable field for exams: **Color**

4. Tap the Settings button or select a different drawer to close the 3D display options.

Using Glass mode

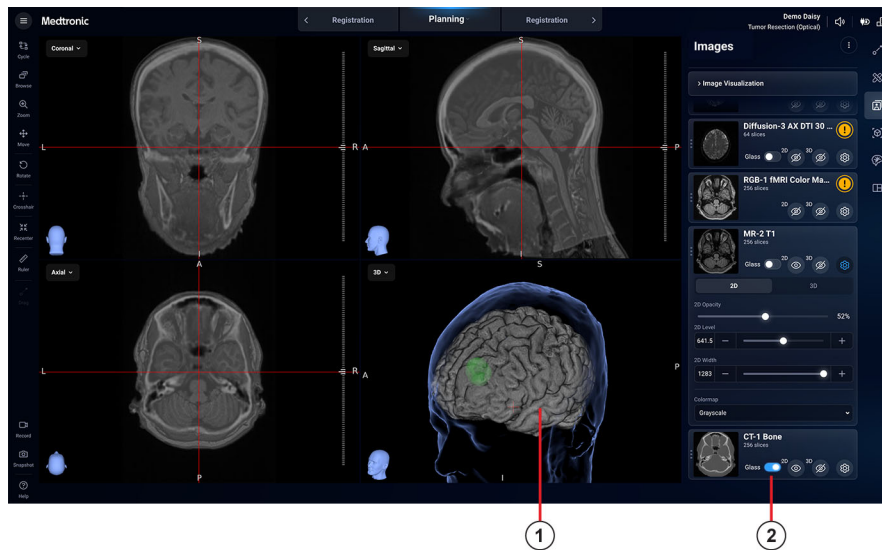
Glass mode allows you to see through outer layers of 3D models, such as skin and bone, to increase the visibility of other models, such as tumor models.

To use Glass mode, follow these instructions:

1. In the **Tractography**, **Planning**, or **Navigation** task, tap the Models button to open the **Models** drawer.
2. Tap the Settings button on a model or image to open its display options.
3. Turn on the **Glass** toggle to enable Glass mode.

In Glass mode, the 3D Opacity is automatically set to 50%. Use the **3D Opacity** slider to adjust the opacity from 0% to 100%.

Note: Glass mode does not change the transparency of fiber tracts.



- 1 Glass mode view
- 2 Glass toggle

Using the 3D Cut mode tool

The 3D Cut mode tool can cut away sections of images and models in the 3D view. To use the 3D Cut mode tool, follow these instructions.

Note: The 3D Cut mode tool does not edit the 3D model. The 3D Cut mode tool only changes the 3D view of the model.

Note: The 3D Cut mode tool does not cut away sections of fiber tracts.

1. In the **Planning** or **Navigation** task, tap the Models button to open the **Images** drawer.
2. Tap **Image Visualization** at the top of the drawer.

3. Turn on the **3D Cut** toggle to enable cut mode.
4. Move the crosshairs to the location you want to see in the 3D view.
The 3D Cut mode tool cuts away a section of the model or image (for which the visibility is currently on) to show the location of the crosshairs in the 3D view.
5. Turn on the **Display 2D Image in Cut Mode** toggle to display the 2D images at the current cut plane.

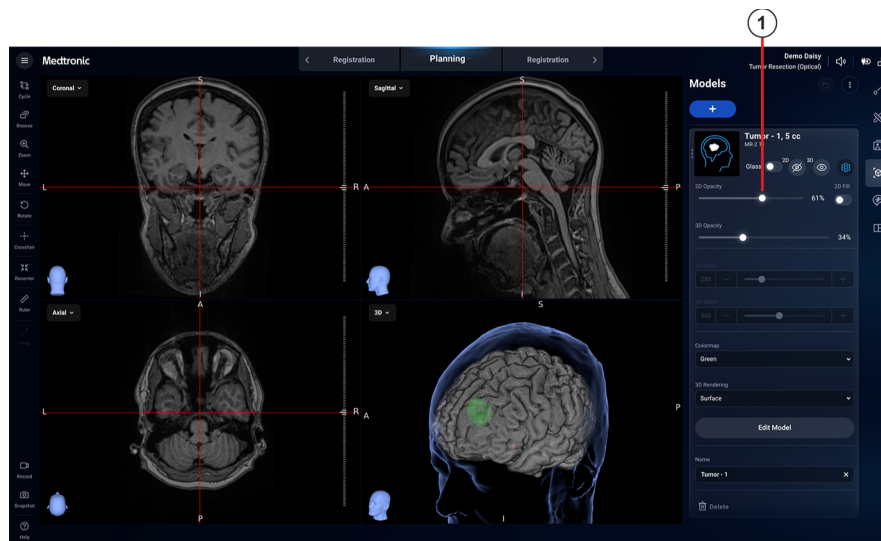
Adjusting the opacity of images and models

In the **Tractography**, **Planning**, and **Navigation** tasks, you can adjust the opacity of images and models.

1. Tap the Images button to open the **Images** drawer or tap the Models button to open the **Models** drawer.
2. Tap the image or model that you want to adjust.
3. Use the **2D Opacity** or **3D Opacity** slider to adjust the image or model's opacity from 0 to 100 percent.

Note: In 2D views, you can re-order image panels to place transparent images on top of opaque images.

Note: For a 2D view, if the **2D Fill** toggle is off, the opacity slider affects the outline of the model. If the **2D Fill** toggle is on, the **2D Opacity** slider affects the model's fill, but not the outline.



1 2D Opacity slider

D: Customizing procedures

You can add customized procedures to a surgeon profile or edit an existing procedure. Customizing procedures allows you to save preferences such as instrument and equipment selection, footswitch behavior, and viewing preferences.

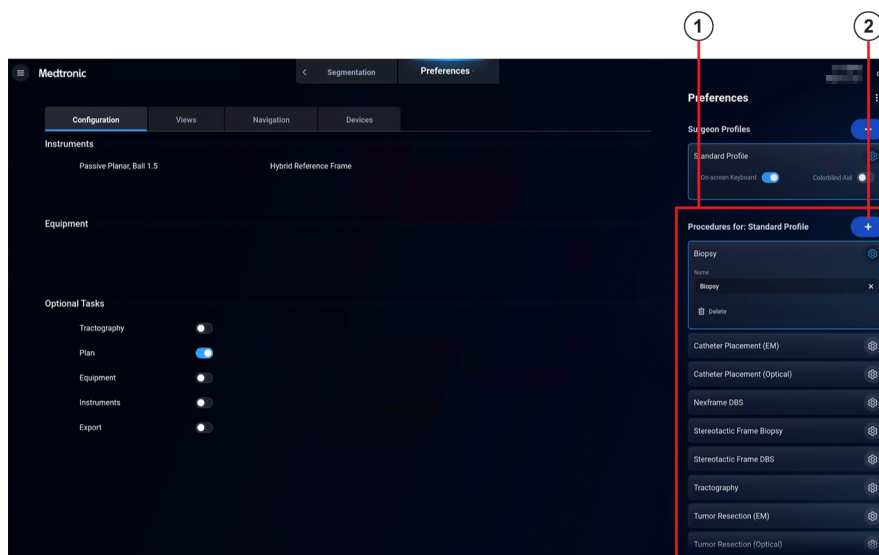
Creating a new procedure

To create a new procedure in your surgeon profile and to customize the procedure selections and surgeon preferences, follow these steps:

Note: To create a new procedure, you must copy an existing procedure or surgeon profile.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile to display a list of available procedures that you can copy.
3. Tap **[+]** next to the procedures list.
4. Tap anywhere in the **Name** field and then enter a name for the procedure.
5. Select a surgeon in the **Copy From** drop-down menu.
6. Select a procedure from the procedure list.
7. Tap **[Create]**.

The procedure is added to the procedure list for the surgeon. All instruments and equipment used in the procedure are automatically saved. You can further edit the procedure to change procedure selections and surgeon preferences. See “Editing a procedure.”



- 1 Procedures list
- 2 [+](Add New Procedure button)

Editing a procedure

Note: Edits made to surgeon profiles only apply to the system the edits are made on.

You can change the procedure selections and surgeon preferences. To edit surgeon preferences for a procedure, follow these steps:

1. Tap the Menu button, and then tap **[Preferences]**.
2. Under **Surgeon Profiles**, select the surgeon profile to display a list of available procedures that you can edit.
3. Select the procedure that you want to edit.

Note: If you want to make an edit during a procedure, the clinical application will automatically edit the surgeon profile and procedure you are currently in.

4. Edit the selections for the behavior and preferences that you want for each tab.

The settings save automatically as you update them.

Exporting surgeon profiles

You can export surgeon profiles when you set up a new system or if you want to update multiple systems with the same surgeon preferences.

Note: If surgeon profiles are imported between a planning station and a workstation cart, only shared settings will update.

1. Tap the Menu button, and then tap **[Preferences]**.
2. Tap the More button, and then tap **[Export Surgeon Profiles]**.
3. Insert media that you want to export surgeon profiles to, and then select it from the **Media** drop-down menu.

Note: You can export surgeon profiles to a USB, CD, or DVD.

4. Tap **[Export]**.
5. Once the export is complete, tap **[Close]**.

Importing surgeon profiles

You can import surgeon profiles when you set up a new system or if you want to update multiple systems with the same surgeon preferences.

Note: If surgeon profiles are imported between a planning station and a workstation cart, only shared settings will update.

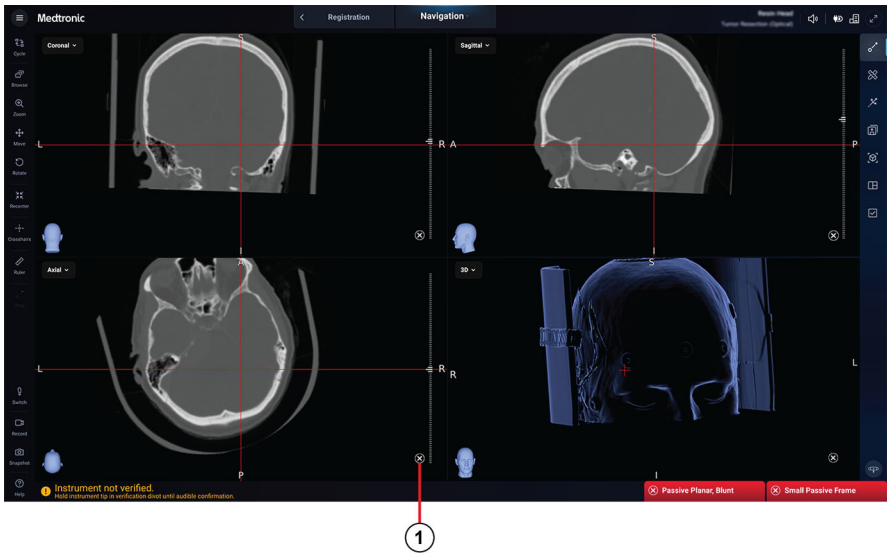
- 1. Tap the Menu button, and then tap **[Preferences]**.
- 2. Tap the More button, and then tap **[Import Surgeon Profiles]**.
- 3. Insert media that contains surgeon profiles you want to import.
Note: You can import surgeon profiles from a USB, CD, or DVD.
- 4. Select the desired surgeon profile file.
- 5. Tap **[Import]**.
- 6. Once the import is complete, tap **[Close]**.

Enabling colorblind aid

The colorblind aid feature allows you to use a symbol in the views to show if the system is tracking a navigated instrument.

- 1. Tap the Menu button, and then tap **[Preferences]**.
- 2. Under **Surgeon Profiles**, select the surgeon profile that you want to edit.
- 3. Tap **Colorblind Aid** to switch it to the on position.

When the colorblind aid is enabled for the surgeon profile, the not navigating indicator displays in the bottom right corner of every image view.



1 The colorblind indicator that the instrument is not navigating.

E: Controls

Touch and mouse interactions

This table describes touch gestures for using the touchscreen to interact with the clinical application and the equivalent mouse interactions. For touch gestures that require only one finger, you can use a capacitive stylus instead of a finger.

Touch gesture	Gesture description	Equivalent mouse interaction	Clinical application action
Tap	Touch and release the screen with one finger.	Click and release the left mouse button	• Initiate an action with a clinical application button • Select a menu option • Identify a location in an image

Touch gesture	Gesture description	Equivalent mouse interaction	Clinical application action
One finger drag	Place a finger on the screen and move it in a direction while maintaining contact with the screen.	Left click and drag	• Rotate the 3D model • Scroll through menu options • Update a sliding value • Use Paint or Erase in Model Building
Two finger drag	Place two fingers on the screen and move them in the same direction while maintaining contact with the screen.	Click [Move] and then click and drag	Move images
Pinch	Place two fingers on the screen and bring them closer together to zoom out.	Right click and drag down	Zoom out on the images
Stretch	Place two fingers on the screen and move them farther apart to zoom in.	Right click and drag up	Zoom in on the images
Two finger rotate	Place two fingers on the screen and rotate one around the other.	Click [Rotate] and then click and drag	Rotate some 2D views
Flick	Quickly slide and release one finger on the screen.	Left click and drag	Swap images in the Images task
Touch and hold	Touch the screen with one finger and hold it in place.	Left click and hold	To use the virtual [Footswitch] instead of a physical footswitch, touch and hold [Footswitch] until you want to release the footswitch.
Double-tap	Touch the screen twice with one finger in rapid succession.	Double-click	• To temporarily change a multiple view layout to a one view layout, double-tap the Image Selection drop-down menu in any view. Double-tap again to return to the previous layout. • Edit a plan name, plan color, or annotation name

Keyboard shortcuts

The following table contains general clinical application keyboard shortcuts.

Keyboard shortcut	Result
<F1>	Selects view layout preset 1, if stored. At the login screen, switches between small and large text sizes.
<F2>	Selects view layout preset 2, if stored. At the login screen, switches between a brighter or darker display.
<F3>	Selects layout view preset 3, if stored. At the login screen, displays or hides the virtual keyboard.
<F4>	Switches between a dot and a crosshair in the 2D views in the Planning , Navigation , or Tractography task.
<F8>	Displays the crosshair image coordinates panel. The coordinates shown are the current crosshair location. Allows user to specify the crosshair position and crosshair trajectory (Hind).
<F9>	Creates a new annotation when the Annotations interface is open.
<Alt> + <t>: applicable for the workstation and planning carts <Alt> + <t>: (Windows) applicable during a remote planning session <Option> + <t>: (Mac) applicable during a remote planning session	Displays all annotation descriptions and saved measurements.
Left arrow or right arrow	Moves crosshairs by 0.1 mm along the current plan when the Plan drawer is open.

Keyboard shortcut	Result
Up arrow or down arrow	Moves crosshairs by 1.0 mm along the current plan when the Plan drawer is open.

System Status Menu

The **System Status Menu** is in the top-right corner of the clinical application screen during a task on the workstation cart or planning station. Select the status icon area to open the **System Status Menu**.

System Status icons	Description
Volume 	You can control the volume by selecting the Volume button. Note: The volume control is not available in remote planning sessions.
Battery Status 	On the workstation cart only, the battery status icon displays the current detected battery status. See “Battery backup” for more detailed battery status descriptions.
Hospital Network 	The hospital network icon indicates whether you are connected to a network by Ethernet or WiFi, or if you are not connected to a network.

Note: The **System Status Menu** does not display in the **Select Procedure** task.

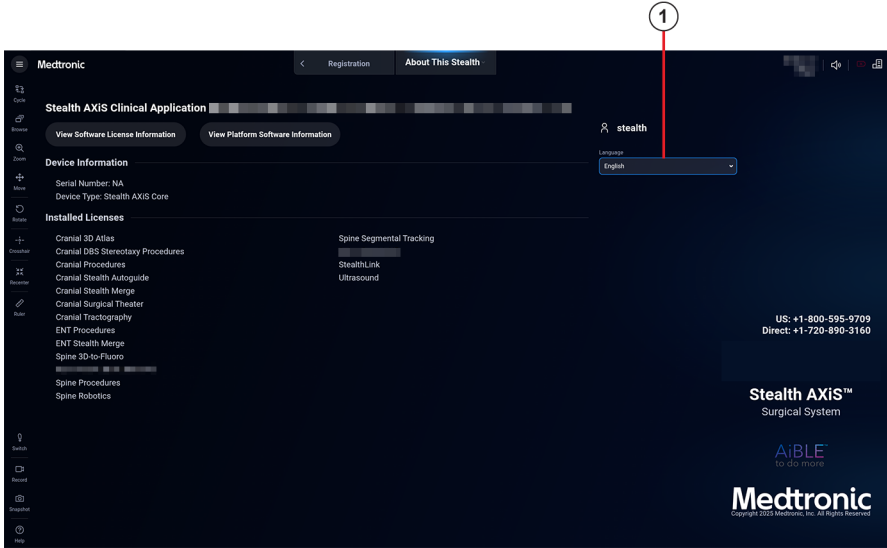
F: Changing the default system setup

Changing the default language

From the **About This Stealth** window you can change the default language.

- 1. From the main menu, tap **About This Stealth**.
- 2. To change the default language settings for the application, tap the **Language** drop-down menu and select the preferred language.

Once a language is selected, the application will immediately update to the selected language.



1 Language drop-down menu

G: DICOM Setup

The **DICOM Setup** task contains networking and DICOM settings for the Stealth AXiS™ System. You can add, edit, and delete PACS servers and other Medtronic Navigation™ systems. You can also run configuration tests and configure a system to act as a PACS server.

Use **DICOM Setup** to complete these tasks:

- Add a PACS server
- Edit a PACS server
- Delete a PACS server
- Change the DICOM settings

To open **DICOM Setup**, follow these steps:

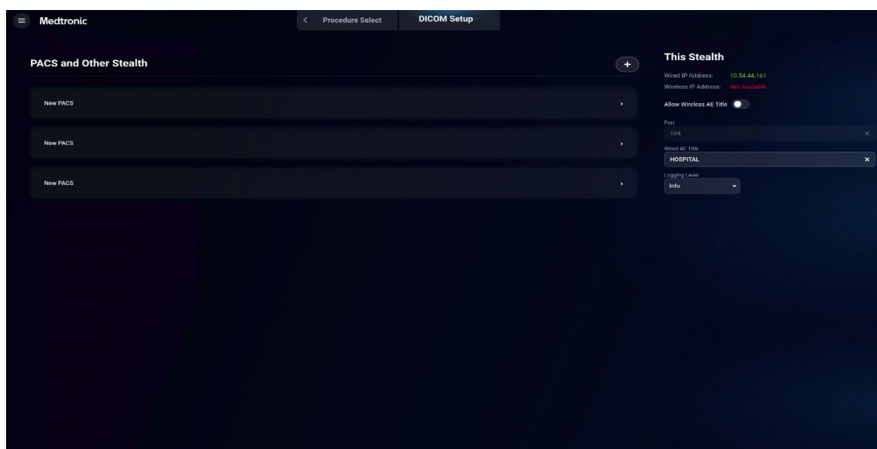
1. Tap the Menu button.
2. Tap **[DICOM Setup]**.

Adding a PACS server

To add a new PACS server, follow these steps:

1. Open the **DICOM Setup** task.
2. Tap **[+]** to add a new PACS server.
3. Enter information for the PACS server:
 - **Name**
 - **AE Title**
 - **IP Address/Hostname**
 - **Port**
4. Select how the system will use the PACS server:
 - **Query**
 - **Storage**
5. To save the PACS server as the default, tap **Default** (optional).

Note: Multiple servers can be configured, but the clinical application can only save one default server at a time.



Editing a PACS server

To edit a PACS server or other Medtronic Navigation™ systems, follow these steps:

1. Open the **DICOM Setup** task.
2. Select an existing DICOM source from the **PACS & Other Stealth** list.
3. You can edit the following:
 - **Name**
 - **AE Title**
 - **IP Address/Hostname**
 - **Port**
4. Select how the system will use the DICOM server:
 - **Query**
 - **Storage**
5. Turn on the **Default** toggle (optional) to save the DICOM server as the default.
Note: Multiple servers can be configured, but the clinical application can only save one default server at a time.

Deleting a PACS server

To delete a PACS server, follow these steps:

1. Open the **DICOM Setup** task.
2. Select an existing DICOM source from the **PACS & Other Stealth** list.
3. Tap **[Delete]**.
4. Tap **[Delete]** on the dialog to confirm. The server will be removed from the list of available servers.

Running DICOM tests

The following DICOM tests can be run:

- Local Config: Displays information about the local configuration.
- Ping: Sends an ICMP echo to the IP address.
- PACS Port: Attempts to open the specified port.
- Association: Tests that the Stealth AXIS™ system can open a DICOM association to the DICOM server.
- DICOM Echo: Performs the C-ECHO DICOM command.

Running all tests

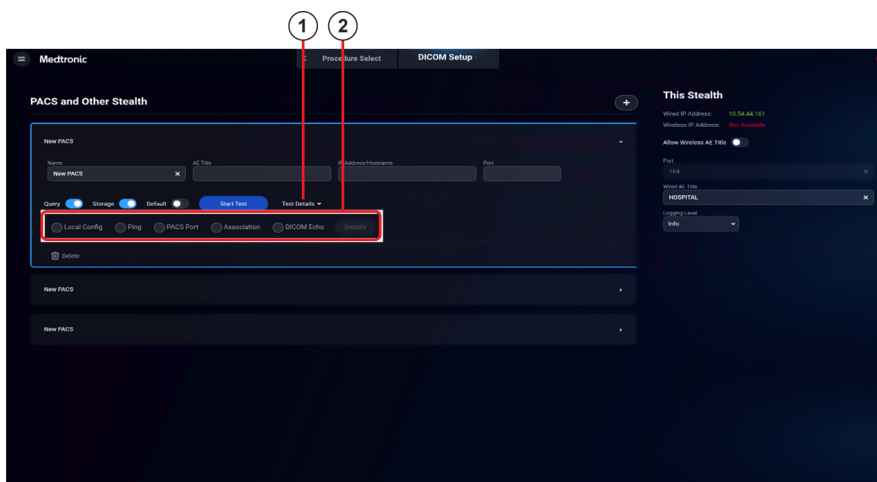
To run all tests and view results:

1. Tap **[Start Test]**.
The **Test Details** drawer opens. As each test is executed, its corresponding button changes to a status color.
 - Green indicates the test passed.
 - Yellow indicates the test encountered a non-fatal error.
 - Red indicates the test failed with a fatal error.
2. Click **[Details]** to view the detailed output of each test run.

Running selected tests

To select which tests you want to run and display the results:

1. Tap the **All Tests** tab.
2. In the **Test Details** drawer, tap the gray button next to each desired test.
3. Tap **[Start Test]**.
As each test is executed, the corresponding gray button changes to a status color.
 - Green indicates the test passed.
 - Yellow indicates the test encountered a non-fatal error.
 - Red indicates the test failed with a fatal error.
4. Click **[Details]** to view the detailed output of each test run.



- 1 All Tests tab
- 2 Test Details drawer

Changing This Stealth DICOM (PACS) settings

Update **This Stealth** DICOM (PACS) settings to configure the system to act like a PACS server for query, storage, or both.

Only one IP address (wired or wireless) can communicate on the network at any given time. Wireless settings are only shown on systems that have wireless network hardware.

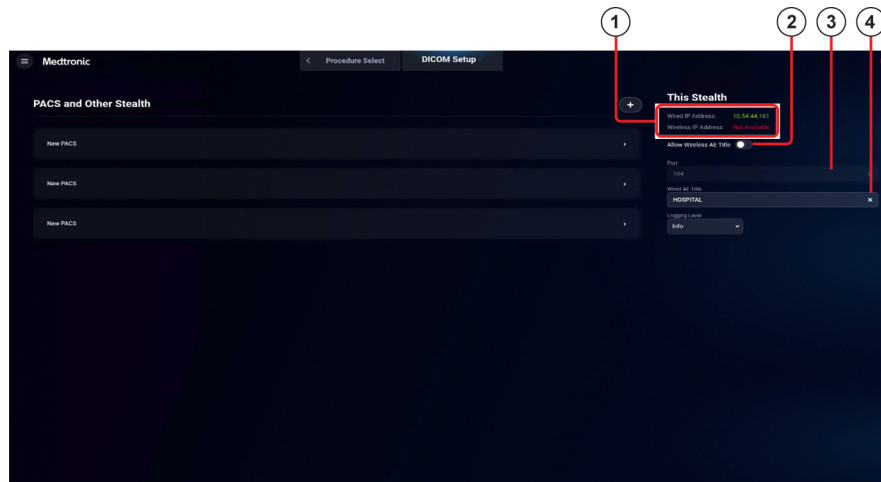
Note: You can use hostnames or IP addresses in the DICOM configuration. These might need to be the fully qualified domain name (FQDN) and not just the hostname by itself, depending on your network setup. Refer to the *Stealth AXIS™ IT Guide* for instructions on how to change IP/hostname.

To enter settings for the system, follow these steps:

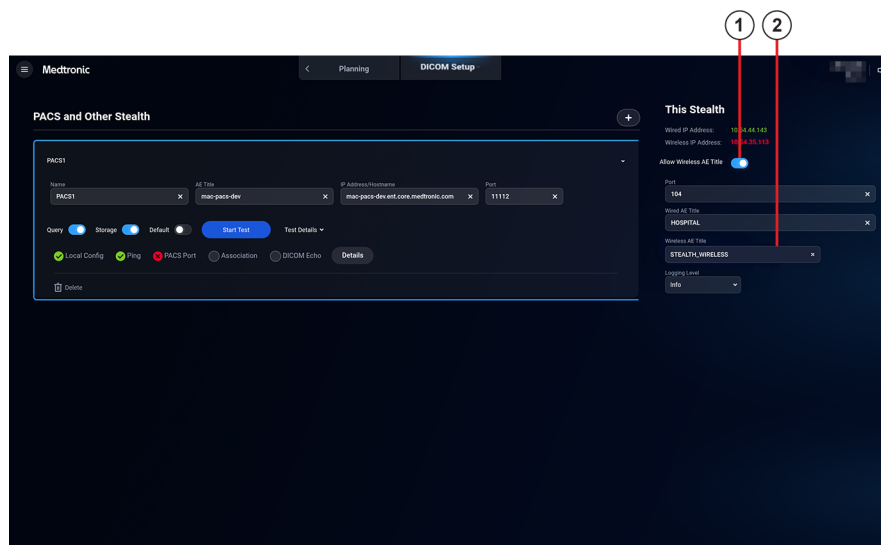
1. Open the **DICOM Setup** task.
2. Change these fields:
 - **Port**
 - **Wired AE Title**

Note: If you are using a workstation cart and the **Allow Wireless AE Title** option is switched on, fields appear for both **Wired AE Title** and **Wireless AE Title**. Both AE title fields can be changed.

Note: Only change the **Logging Level** from its default setting, "Info", at the request of Medtronic technical services.



- 1 If the system is able to communicate on the network, the IP address label is green. If the system is not able to communicate, the label is red. Only one IP address (wired or wireless) can be green at any given time.
- 2 Wireless settings are only shown on systems that have wireless network hardware.
- 3 **Port** field
- 4 **Wired AE Title** field



- 1 **Allow Wireless AE Title** toggle turned on
- 2 **Wireless AE Title** field

H: Buttons

Buttons in Images, Tractography, or Planning

The following buttons display in the **Images**, **Tractography**, or **Planning** tasks.

	[Cycle] cycles through the preset image layouts.		[Move] allows you to move an image within a view. Tap [Move] and then drag an image.
	[Zoom] magnifies an image. Tap [Zoom] and drag up or down on an image.		[Recenter] returns the crosshairs to the centers of the images.
	[Correct] displays options to reorient the images.		[Snapshot] captures a snapshot of the screen and stores it in the patient data.
	[Browse] scrolls through image slices. Tap this button and then drag up or down on an image.		[Help] displays relevant help information.
	[Record] captures and records a video of the screen and stores it in the patient data.		[Rotate] allows you to rotate the 3D model and the trajectory views. Drag the 3D model in the direction that you want it to rotate. In a trajectory view, tap one of the rotation buttons to turn the image 90°.
	[Crosshairs] switches between crosshairs and dots in the 2D views.		[Drag] allows you to drag the plan entry or target point. This enables you to update the entry and plan length.
	The layout buttons change the screen layout to the preset image views.		[Ruler] displays the distance in millimeters between any two selected points on an image.
	The Image Selection drop-down menu allows you to select the content in each view. Tap this button and choose from the drop-down list (for example, Axial , Coronal , Sagittal , 3D , or Video).		

I: Image label translations

The clinical application displays image orientation labels on the edges of the image views to help identify the patient's orientation. The orientation labels are the initials for the six main anatomical location terms: inferior, superior, anterior, posterior, left, and right.

The image orientation labels may not be translated in the clinical application. They are labeled by the English initials I, S, A, P, L, R, and L.

Label	Meaning
I	Inferior
S	Superior
A	Anterior
P	Posterior
L	Left
R	Right



J: Symbols

Explanation of symbols on package labeling

The following symbols may appear on system equipment, on system packaging, on accessories used with the system, or in this document.



Medical Device



cTUVus Mark certifies the product meets the requirements of a Nationally Recognized Testing Laboratory for compliance with IEC 60601-1 and CAN/CSA-C22.2 No. 60601-1:14.



Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.



Caution



Follow instructions for use.



Consult instructions for use.



Consult electronic instructions for use.



Type BF applied part, in compliance with IEC/UL60601-1.



Type B applied part, in compliance with IEC/UL60601-1.



Single use only. Do not reuse.



Use by date



Batch code



Serial number



Catalog number



Quantity

	Sterile
	Sterilized using ethylene oxide
	Sterilized using irradiation
	Single sterile barrier system
	Single sterile barrier system with protective packaging inside
	Single sterile barrier system with protective packaging outside
	Double sterile barrier system
	Non-sterile
	Do not sterilize.
	Do not resterilize.
	Manufacturer
	Authorized representative in the European Community
	Responsible person in Great Britain (England, Wales, and Scotland)
	Importer
	Date of manufacture
	Keep dry.
	Do not use if package is damaged.
	Fragile, handle with care.
	Keep upright.
	Do not stack.
	Power on. Connect to main power.
	Power off. Disconnect from main power.
	Power on for part of the system.
	Power off for part of the system.
	Power button
	Protective Earth (ground)
	AC Power



Equipotentiality: identifies the terminal that when connected together, bring the various parts of the equipment or system to the same potential, not necessarily being earth potential (for local bonding).



Dead CMOS battery



Do not allow contact with patient. Temperature may exceed limits.



Atmospheric pressure limits



Humidity limits



Temperature limits



Localizer must not be used in ambient temperatures greater than 30°C (86°F).



Storage temperature between -29°C and 60°C (-20°F and 140°F)



Shipping temperature between -30°C and 60°C (-22°F and 140°F)



MR Conditional. Certain accessories that may be used with the system are MR Conditional. See the labeling for those devices for additional instructions on MR conditional use.



Do not disassemble.



USB 2.0 port



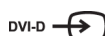
USB port: Connects the system to an external high-speed USB device.



Ethernet port: Connects the system to the site network and medical imaging devices.



DisplayPort for video out



DVI-D port for video in



Footswitch



Camera cart port: Connects the camera cart to the workstation cart.



Robot cart port: Connects the robot cart to the workstation cart.



Workstation cart port: Connects the workstation cart to the robot cart.



Cart docking location



EM instrument interface



EM emitter



Microscope: Connects the system to a surgical microscope. The orange port is for analog microscopes, and the yellow port is for digital microscopes.



SCU ports: Connects an active, wired instrument (patient reference frame or microscope tracker) to the system.



Radio frequency device. Interference may occur in the vicinity of the device.



Protect from heat and radioactive sources

IPX0

Not protected

IPX2

Protected against dripping water when tilted up to 15 degrees

IPX6

Protected against powerful water jets



Locked



Unlocked



Motion lock



Directional lock



Robot arm protective stop



Do not transport the camera cart with a camera or monitor in an undocked position. Always dock the camera and monitor before moving the camera cart. Use the handle to push the cart.



Do not transport the workstation cart with the monitor in an undocked position. Always dock the monitor and close drawers before moving the workstation cart. Use the handle to push the cart.



Do not push the cart from the back of the cart or from the labeled or adjacent surfaces.



LASER radiation emitted from aperture. Do not stare into beam. Class 2 LASER product. Maximum output: <1 mW, CW: 635–670 nm, IEC/EN 60825-1 (2014), ANSI Z136.1 (2014). Complies with 21 CFR 1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3., as described in Laser Notice No. 56, dated May 8, 2019.



Do not dispose of this product in the unsorted municipal waste stream. Dispose of this product according to local regulations. See <http://recycling.medtronic.com> for instructions on proper disposal of this product.



China RoHS compliant. Environmental protection use period of 5 years. Environmental protection use period of 10 years. Environmental protection use period of 50 years.



China RoHS compliant: Electronic Information Products Pollution Control symbol. (The “e” in the logo signifies none of the restricted substances are present above permitted levels.)



Open here

R_x Only

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